

Engineering Documentation for Conservation of Wat Phra Mahathat Woramahawihan, Nakhon Si Thammarat



This study is supported by
Thailand Science Research and Innovation
29 June 2019

Engineering Documentation for Conservation of Wat Phra Mahathat Woramahawihan, Nakhon Si Thammarat

Prepared by

Nakhorn Poovarodom	Thammasat University
Naret Limsamphancharoen	Thammasat University
Krisada Chaiyasarn	Thammasat University
Sutat Leelataviwat	King Mongkut's University of Technology Thonburi
Chainarong Athisakul	King Mongkut's University of Technology Thonburi

This research is supported by Thailand Science Research and Innovation

29 June 2019

TABLE OF CONTENTS

	Page
Chapter 1 INTRODUCTION.....	1-1
Chapter 2 3D DIGITAL DOCUMENTATION WITH TERRESTRIAL LASER SCANNING.....	2-1
2.1 Introduction.....	2-1
2.2 3D Terrestrial Laser Scanning Technology.....	2-1
2.3 3D point cloud data process.....	2-2
2.4 Key Plans and Grid Layout.....	2-4
2.5 Data Depository.....	2-15
Chapter 3 3D DIGITAL DOCUMENTATION WITH IMAGE-BASED 3D MODELING.....	3-1
3.1 Introduction.....	3-1
3.2 Data acquisition.....	3-1
3.3 Registration.....	3-3
3.4 Deformation Assessment.....	3-4
3.5 SfM to solid modeling.....	3-6
3.6 A UAV based inspection for ancient structure.....	3-9
Chapter 4 IDENTIFICATION OF NATURAL FREQUENCY STRUCTURE BY AMBIENT VIBRATION TEST.....	4-1
4.1 Fundamental of the Identification Technique.....	4-1
4.2 Measuring equipment.....	4-2
4.3 Ambient Vibration Tests of Wat Phra Mahathat Woramahawihan.....	4-2
4.4 Results of Chedi Phra Mahathat.....	4-3
Chapter 5 FINITE ELEMENT MODELING OF CHEDI PHRA MAHATHAT.....	5-1
5.1 Geometric modeling and finite element discretization.....	5-1
5.2 Material properties.....	5-3
5.3 Stress analysis under self-weight.....	5-4
5.4 Modal dynamic properties.....	5-7
5.5 Stress analysis under self-weight of tilting model.....	5-8
5.6 Summary.....	5-10
Chapter 6 ASSESSMENT OF TRAFFIC INDUCED VIBRATION.....	6-1
6.1 Introduction.....	6-1
6.2 Method of study.....	6-1
6.3 Site condition.....	6-3
6.4 Attenuation of Vibration Wave.....	6-4
6.5 Vibration Intensity and Acceptance Criteria for Historic Buildings.....	6-1
Chapter 7 SUMMARY.....	7-1



LIST OF TABLES

	Page
Table 2.1: Performance specifications of the FARO Laser Scanner Focus3D X330 used in this project.	2-1
Table 2.2: The height at each key location of the main structure	2-10
Table 3.1: A summary table of the registration result between the Agisoft and LIDAR models.	3-3
Table 4.1: Natural period of the fundamental mode of the investigated chedis.....	4-7
Table 5.1: Modal frequency and effective mass.....	5-8

LIST OF FIGURES

	Page
Figure 2.1: 3D Terrestrial laser scanning on the actual site.	2-1
Figure 2.2: 3D point clouds of Wat Phra Mahathat Woramahawihan.	2-2
Figure 2.3: View of Wat Phra Mahathat Woramahawihan (North Direction).....	2-3
Figure 2.4: View of Wat Phra Mahathat Woramahawihan (South Direction).....	2-3
Figure 2.5: View of Wat Phra Mahathat Woramahawihan (East Direction).....	2-4
Figure 2.6: View of Wat Phra Mahathat Woramahawihan (West Direction).....	2-4
Figure 2.7: The current plan view of Wat Phra Mahathat Woramahawihan.....	2-5
Figure 2.8: Elevation view of Wat Phra Mahathat Woramahawihan (North direction).....	2-6
Figure 2.9: Elevation view of Wat Phra Mahathat Woramahawihan (South direction)	2-6
Figure 2.10: Elevation view of Wat Phra Mahathat Woramahawihan (East direction).....	2-7
Figure 2.11: Elevation view of Wat Phra Mahathat Woramahawihan (West direction).....	2-7
Figure 2.12: Elevation contour of the temple area.....	2-8
Figure 2.13: Elevation contour of the surrounding area.....	2-9
Figure 2.14: Different sections along the height of Wat Phra Mahathat Woramahawihan main structure.....	2-9
Figure 2.15: Inclination direction of Chedi Wat Phra Mahathat Woramahawihan.....	2-10
Figure 2.16: Elevation view of Chedi Wat Phra Mahathat Woramahawihan.....	2-11
Figure 2.17: Section A-A.....	2-12
Figure 2.18: Section B-B.....	2-12
Figure 2.19: Section C-C.....	2-13
Figure 2.20: Section D-D.....	2-13
Figure 2.21: Section E-E.....	2-14
Figure 2.22: Section F-F.....	2-14
Figure 2.23: Section G-G.....	2-15
Figure 2.24: Webshare of Point Cloud Data.....	2-15
Figure 2.25: Webshare of Point Cloud Data.....	2-16
Figure 2.26: Webshare of Point Cloud Data.....	2-16
Figure 2.27: Webshare of Point Cloud Data.....	2-17
Figure 3.1: Top view image from UAV of the site.....	3-1
Figure 3.2: A 3D model of the temple created from Agisoft with camera positions.....	3-2
Figure 3.3: (a) 3D model from Agisoft Photoscan (b) 3D model from LIDAR.....	3-3
Figure 3.4: The point cloud registration between the models from Agisoft and LIDAR.....	3-4

Figure 3.5: (a) Show the location of the center of the temple; (b) Angles of inclination of the top and bottom parts of the temple.....	3-5
Figure 3.6: Results of inclined angles on the top part (a) South direction and (b) East direction	3-5
Figure 3.7: Results of inclined angles of the structure (a) South direction and East direction ..	3-6
Figure 3.8: Flight path of small Chedi in zone 1	3-6
Figure 3.9: Converting 3D model, (a) mesh model (b) Solid model	3-7
Figure 3.10: 3D model of chedi in zone 1	3-7
Figure 3.11: 3D model of chedi in zone 2	3-8
Figure 3.12: 3D model of chedi in zone 3	3-8
Figure 3.13: Converting 3D model, (a) point cloud (b) Solid model	3-9
Figure 3.14: Surface from reconstructed model	3-10
Figure 3.15: Surface from reconstructed model	3-10
Figure 4.1: Velocity measuring system	4-2
Figure 4.2: Area of the chedis selected for ambient vibration test	4-3
Figure 4.3: Locations of vibration sensors for ambient vibration test of Chedi Phra Mahathat	4-4
Figure 4.4: Example of ambient vibration signal	4-5
Figure 4.5: Fourier spectra plots (from top; level F4, F3, F2 and F1)	4-5
Figure 4.6: The chedis for measurement in zone 1	4-6
Figure 4.7: The chedis for measurement in zone 2 and 3	4-6
Figure 4.8: Relations of the natural period and height of the subsidiary chedis	4-8
Figure 5.1: Plan view of Chedi Phra Mahathat model	5-1
Figure 5.2: Side view of Chedi Phra Mahathat model	5-2
Figure 5.3: Three-dimensional view (left) and section 1-1 (right)	5-2
Figure 5.4: Finite element mesh	5-3
Figure 5.5: Maximum principal stress	5-4
Figure 5.6: Minimum principal stress	5-5
Figure 5.7: Distribution of compressive	5-5
Figure 5.8: Distribution of safety factor under self-weight	5-6
Figure 5.9: Distribution of bearing stress beneath the base	5-7
Figure 5.10: Distribution of bearing stress along N-S	5-7
Figure 5.11: First three vibration modes in x-di	5-8
Figure 5.12: Tilting of the model	5-9
Figure 5.13: Effect of model's tilting on safety factor	5-10

Figure 6.1: Location of the vibration sensors and Rajdamnern road.....	6-2
Figure 6.2: Arrangement of vibration sensor (1).....	6-2
Figure 6.3: Arrangement of vibration sensor (2).....	6-2
Figure 6.4: Arrangement of sensor for microtremor array measurement.....	6-3
Figure 6.5: (a) Dispersion curve and (b) Shear wave velocity profile at the site.....	6-4
Figure 6.6: Attenuation of vibration amplitude between two sensors.....	6-5
Figure 6.7: Attenuation of vibration and distance from source (1).....	6-5
Figure 6.8: Attenuation of vibration and distance from source (2).....	6-6
Figure 6.9: Peak particle velocity for all sensors.....	6-8
Figure 6.10: Occurrence rate per hour of the peak velocity at sensor number 1.....	6-8

Chapter 1 INTRODUCTION

Wat Phra Mahathat Woramahawihan, Nakhon Si Thammarat is one of the most valuable historical remains in Thailand. This heritage site has possible threats as the appearance and structural conditions have been deteriorating due to aging, environmental impacts and man-made activities. Proper maintenance and monitoring plan is therefore very important for preservation of the site. In order to have rational plan and decision, documentations and assessment of the existing conditions of the structures are necessary. In this research, the main objective is to apply engineering measures for documentations of the existing condition of Wat Phra Mahathat Woramahawihan. The obtained results will support monitoring schemes for the site to achieve a goal of sustainable preservation.

The study begins with geometry survey of the structures using 3-dimensional digital documentation techniques. Two advanced techniques are employed in the study; the terrestrial laser scanning technique, as detailed in Chapter 2 and the image-based modeling technique, as detailed in Chapter 3. Dynamic tests of the chedis to identify their natural frequencies are explained in Chapter 4. Then, structural analysis of Chedi Phra Mahathat is performed in Chapter 5 to examine the stability of the structure using 3D Finite Element Modelling constructed from the results in Chapter 2 to 4. Finally, effects of traffic induced vibration are examined in Chapter 6.

Chapter 2 3D DIGITAL DOCUMENTATION WITH TERRESTRIAL LASER SCANNING

2.1 Introduction

For the purpose of 3D digital documentation, terrestrial laser scanning technology was used to collect point cloud data for Wat Phra Mahathat Woramahawihan. This section describes the details of the documentation process.

2.2 3D Terrestrial Laser Scanning Technology

The FARO Laser Scanner Focus3D X 330 was used to collect the three-dimensional point cloud data of Wat Phra Mahathat Woramahawihan (Figure. 2.1) with millimeter precision. This laser scanner uses the phase shift technology to measure the distance together with the vertical and horizontal angles. These angles and distances are transformed to the Cartesian coordinate (X,Y,Z). The performance specifications of the scanner used for this project are listed in Table 2.1.



Figure 2.1: 3D Terrestrial laser scanning on the actual site.

2.3 3D point cloud data process

For this project, seventy-four laser scanner stations were used to collect the point cloud data of the area. Using these seventy-four laser scanner stations, approximately one billion 3D point clouds of the Wat Phra Mahathat Woramahawihan were generated as represented in Figure. 2.2-2.6.



Figure 2.1: 3D point clouds of Wat Phra Mahathat Woramahawihan.



Figure 2.2: View of Wat Phra Mahathat Woramahawihan (North Direction)



Figure 2.3: View of Wat Phra Mahathat Woramahawihan (South Direction)



Figure 2.4: View of Wat Phra Mahathat Woramahawihan (East Direction)



Figure 2.5: View of Wat Phra Mahathat Woramahawihan (West Direction)

2.4 Key Plans and Grid Layout

Key plan and grid lines were generated for the area around the Wat Phra Maha Wat Woramahawihan for the purposes of documentation. The key plan and the grid lines are shown in Figure. 2.7 - 2.11. In addition, the contour plot showing the relative elevation values of the surrounding area was also generated as shown in Figure 2.12. Cross-sectional views are provided in Figure. 2.12 - 2.23. The generated point cloud data can give a valuable contribution to future structural monitoring and analysis by providing information about displacements describing tilt, shift, slide, twist, etc.

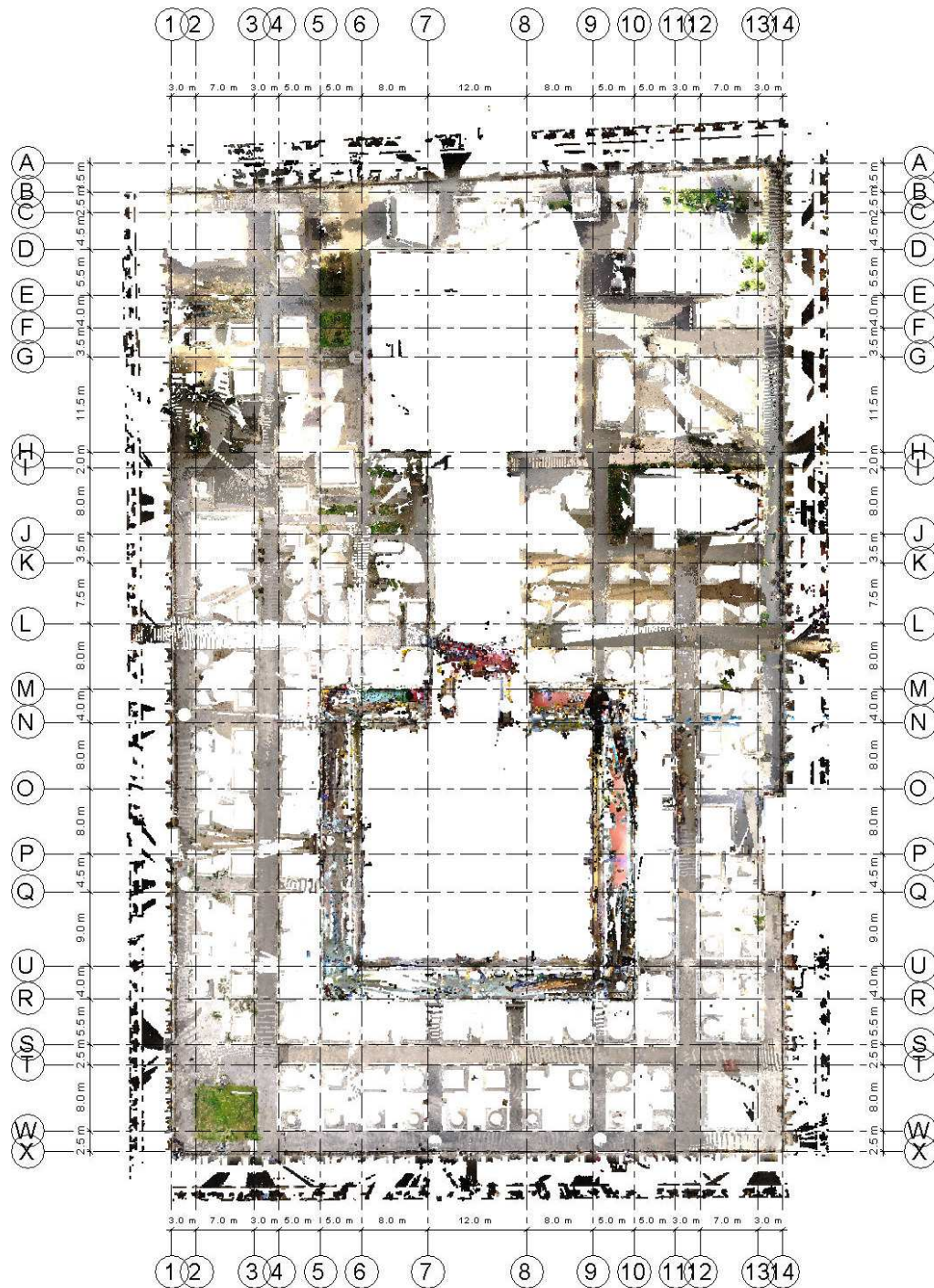


Figure 2.6: The current plan view of Wat Phra Mahathat Woramahawihan

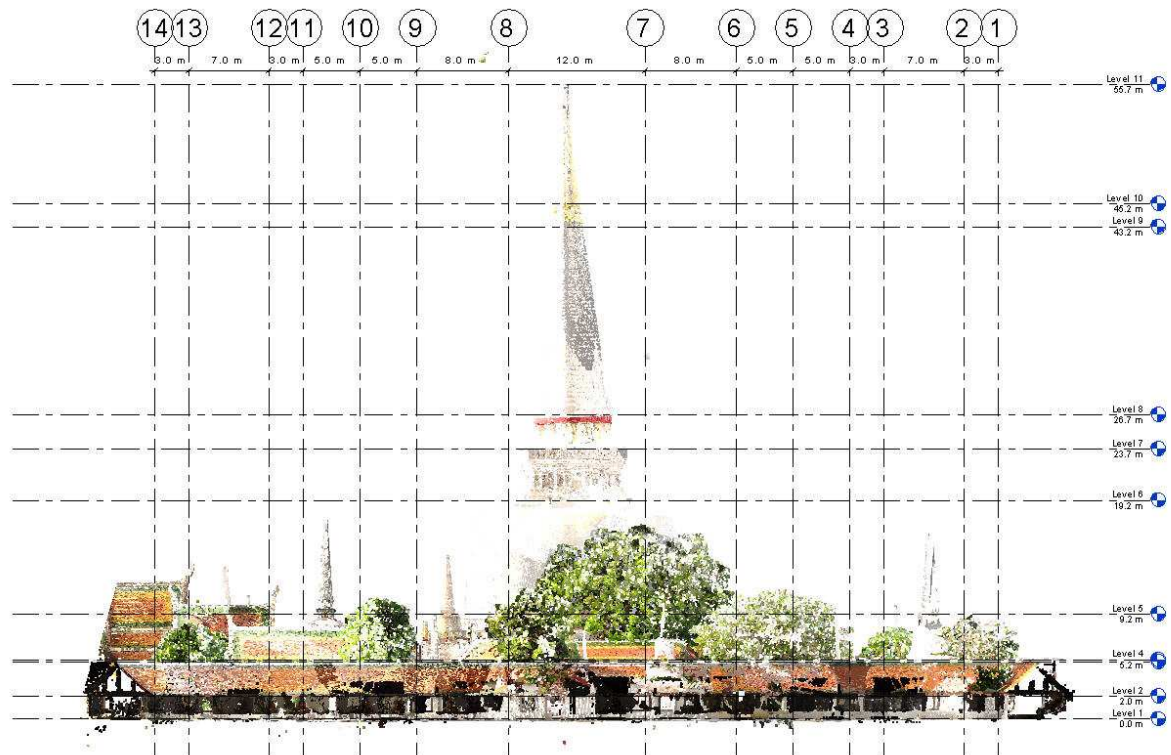


Figure 2.7: Elevation view of Wat Phra Mahathat Woramahawihan (North direction)

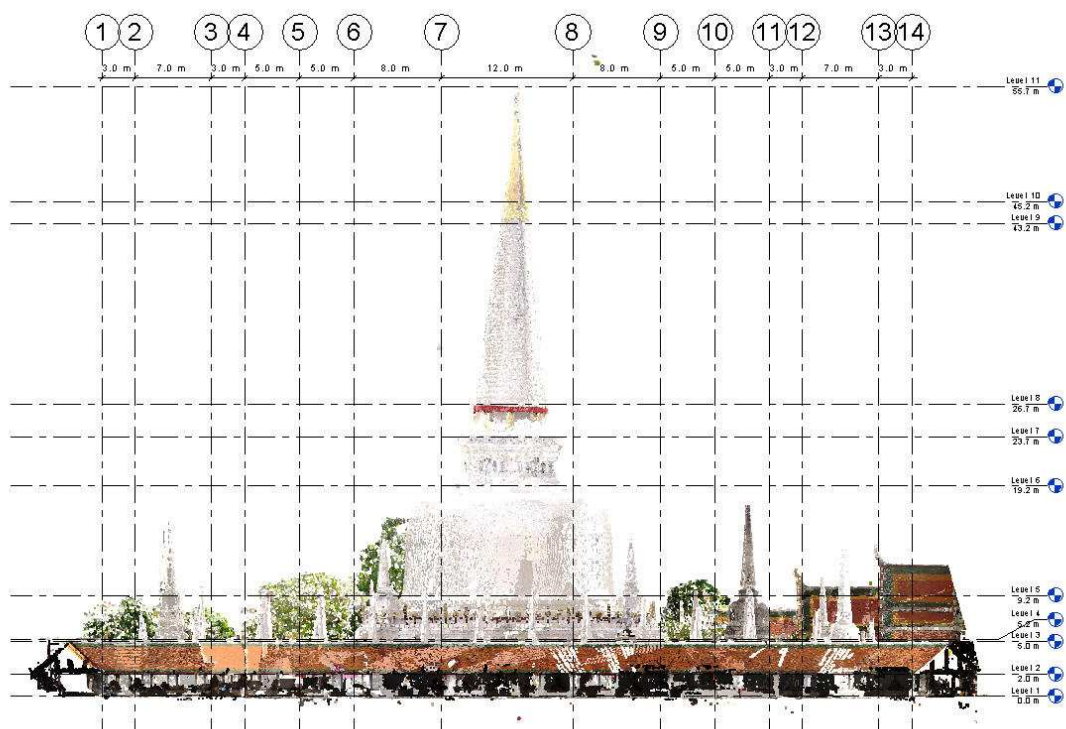


Figure 2.8: Elevation view of Wat Phra Mahathat Woramahawihan (South direction)

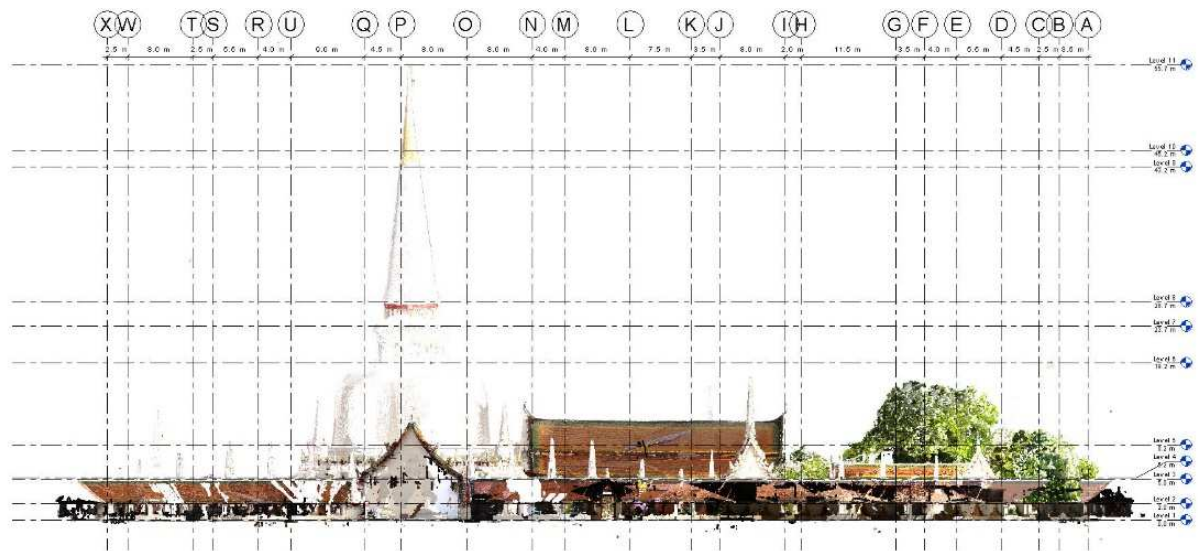


Figure 2.9: Elevation view of Wat Phra Mahathat Woramahawihan (East direction)

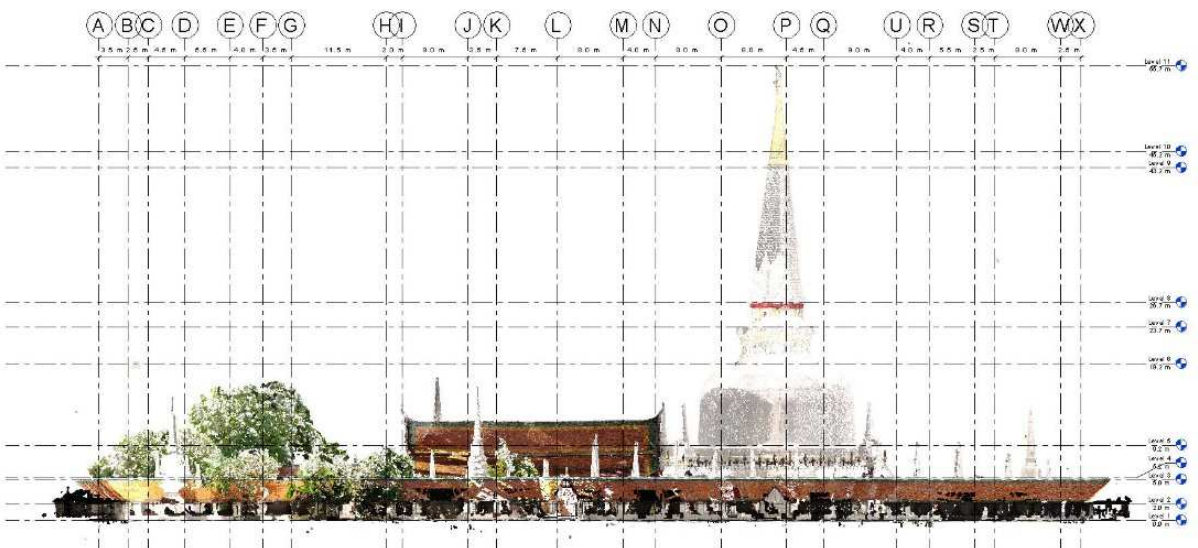


Figure 2.10: Elevation view of Wat Phra Mahathat Woramahawihan (West direction)

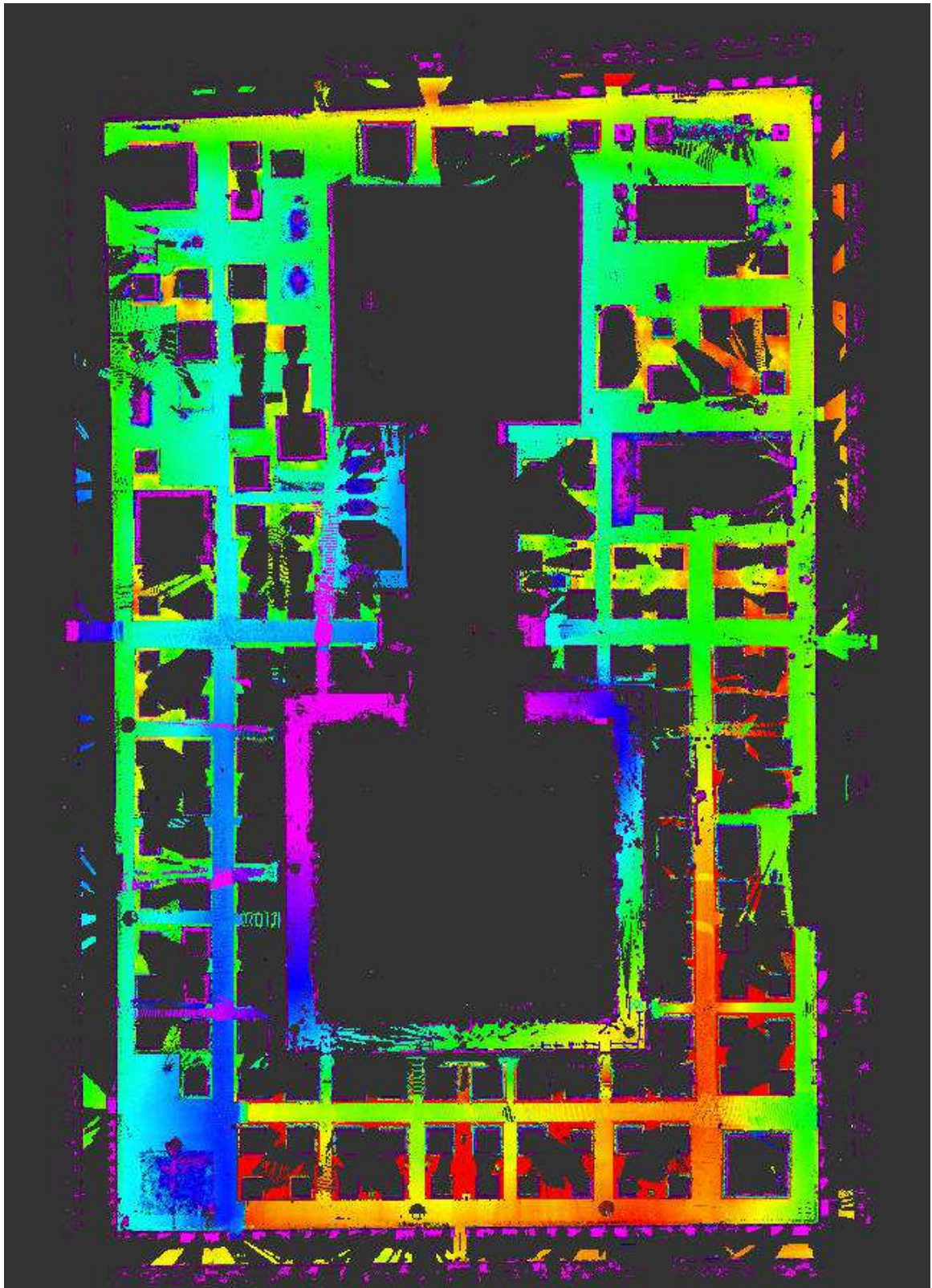


Figure 2.11: Elevation contour of the temple area

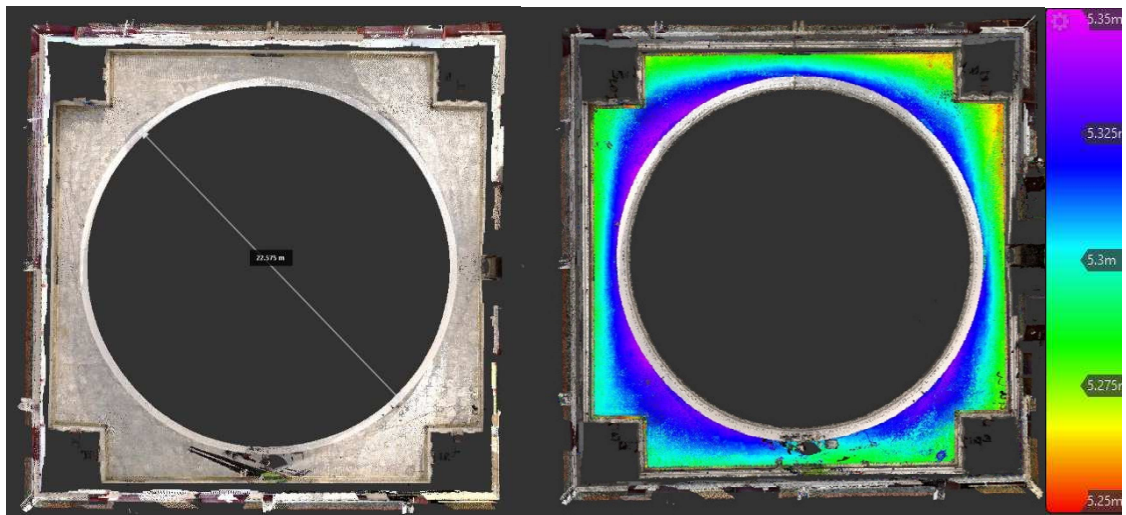


Figure 2.12: Elevation contour of the surrounding area

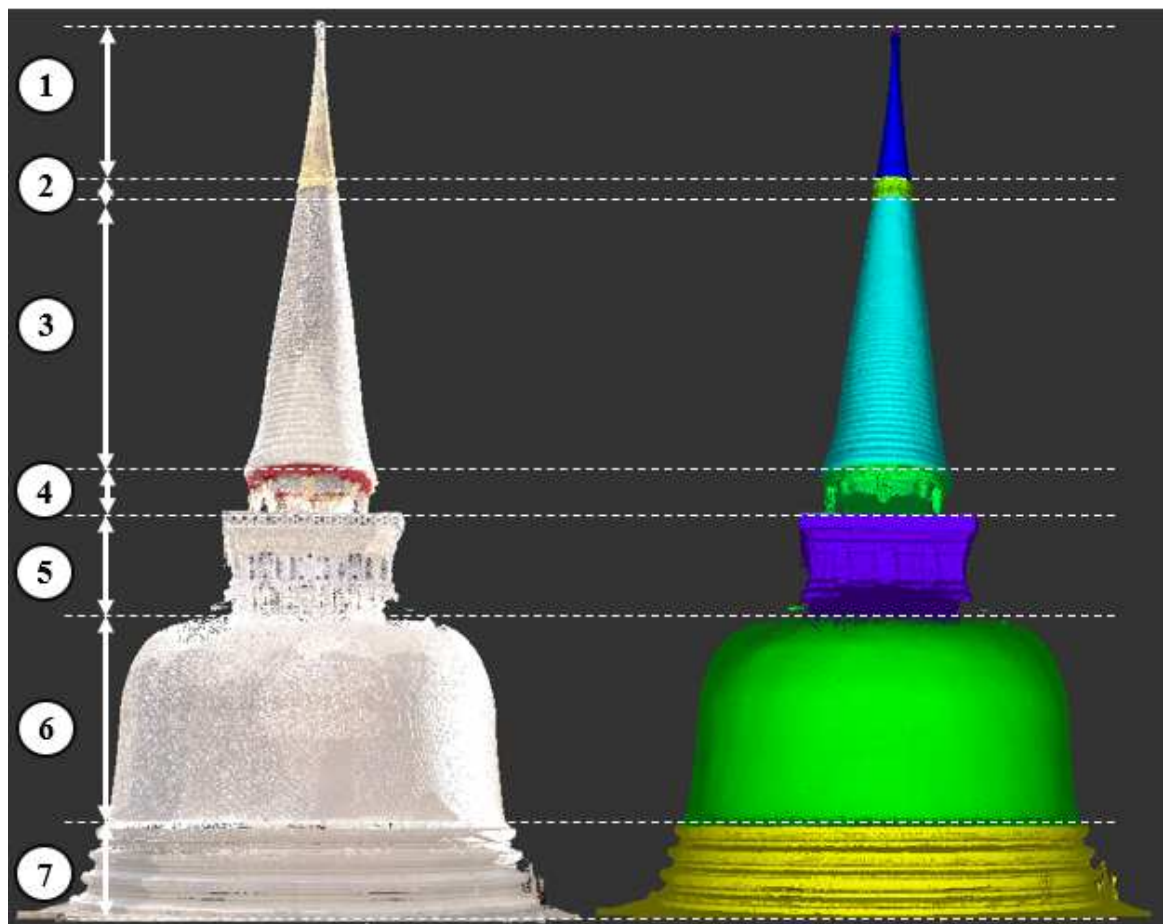


Figure 2.13: Different sections along the height of Wat Phra Mahathat Woramahawihan main structure.

Table 2.1: The height at each key location of the main structure

Section Number	Height (Meter)
1	10.784
2	1.525
3	16.512
4	2.104
5	4.498
6	10.273
7	3.341

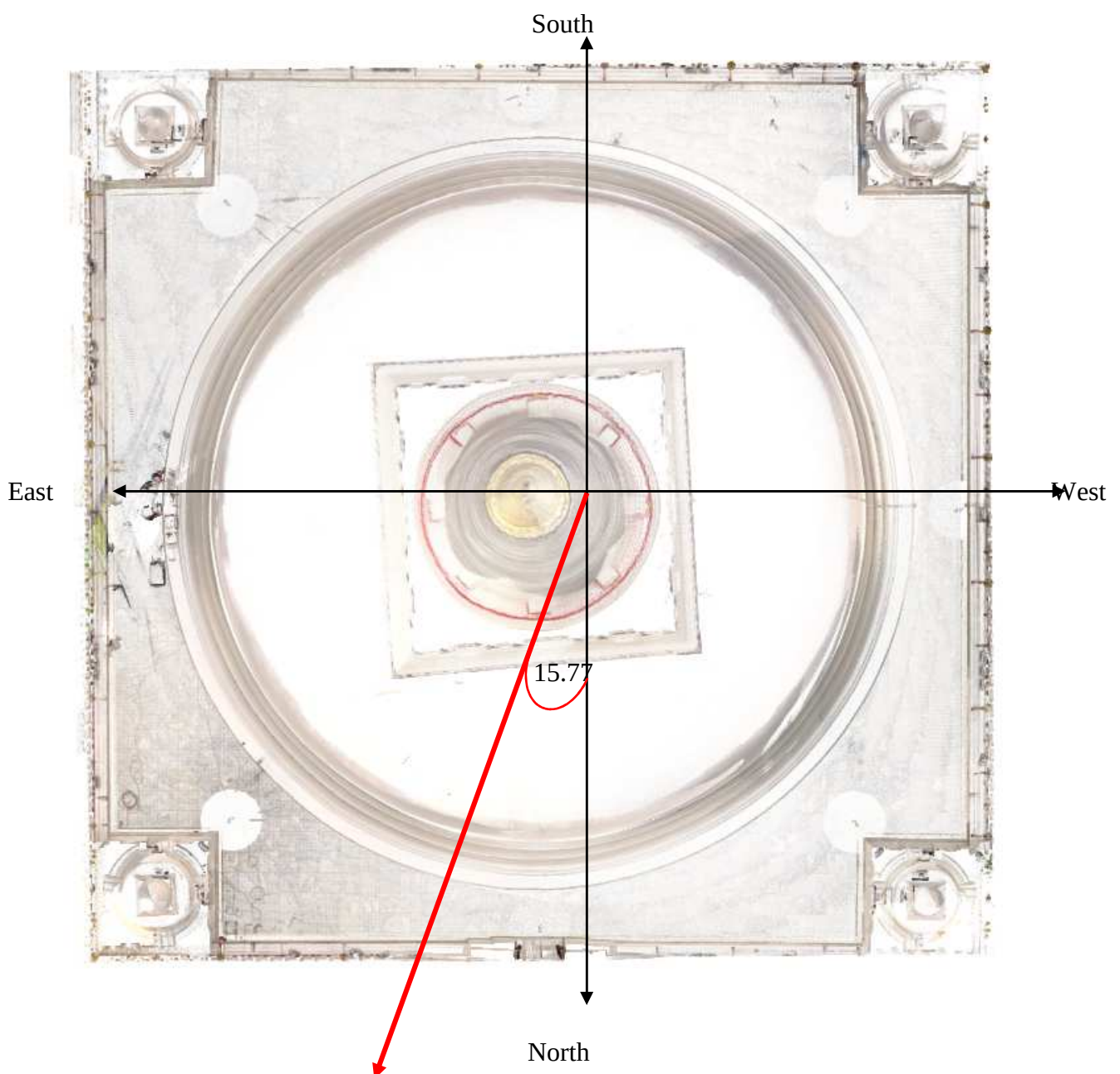


Figure 2.14: Inclination direction of Chedi Wat Phra Mahathat Woramahawihan.

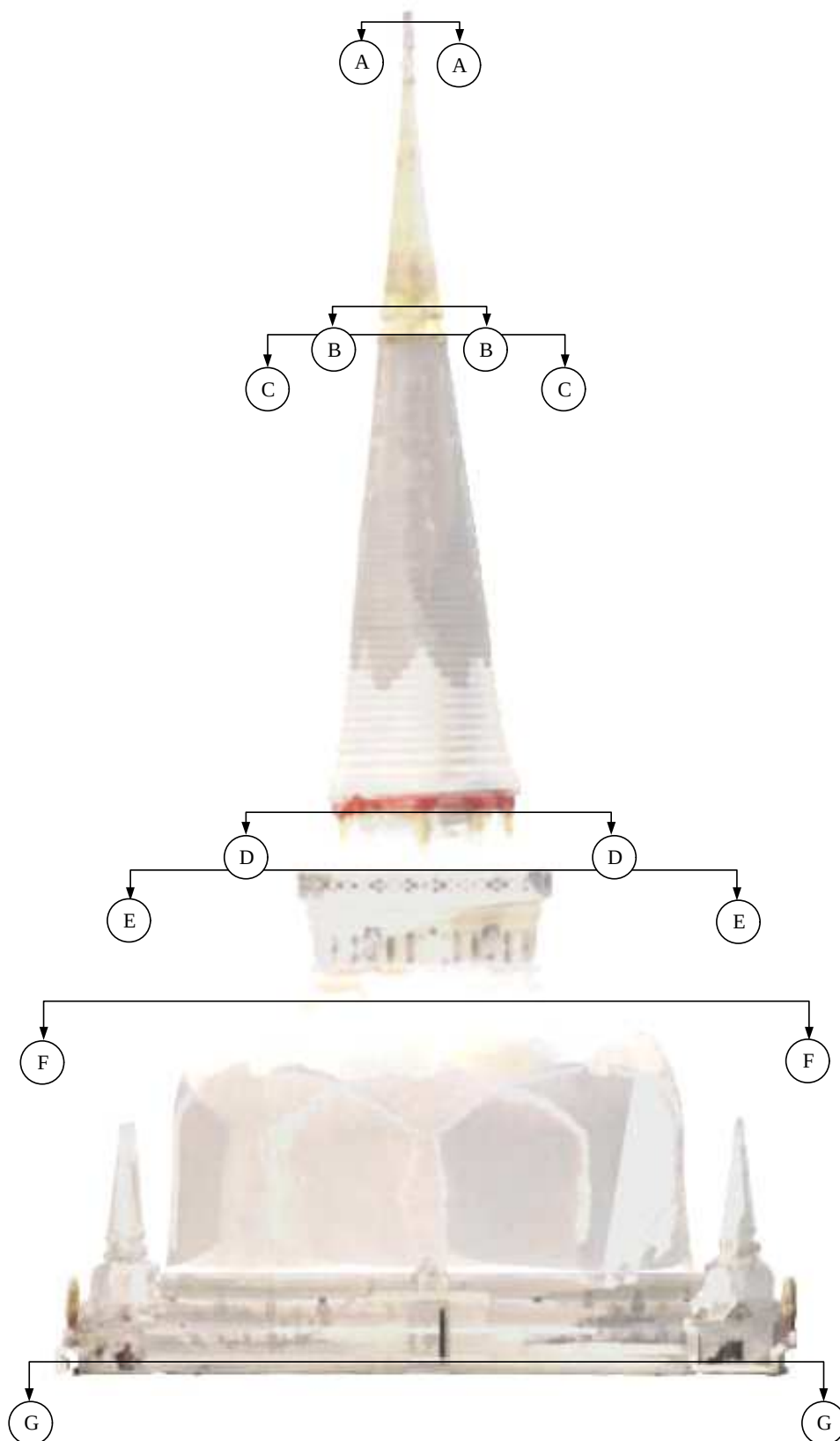


Figure 2.15: Elevation view of Chedi Wat Phra Mahathat Woramahawihan

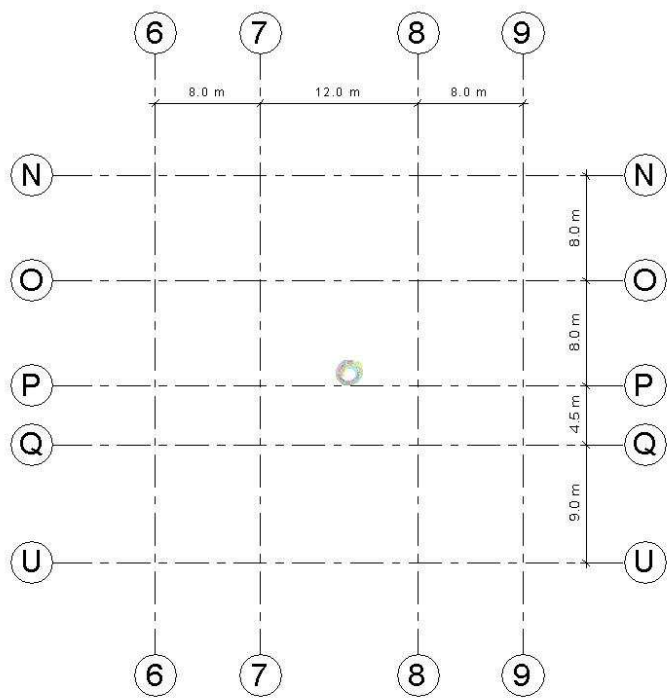


Figure 2.16: Section A-A

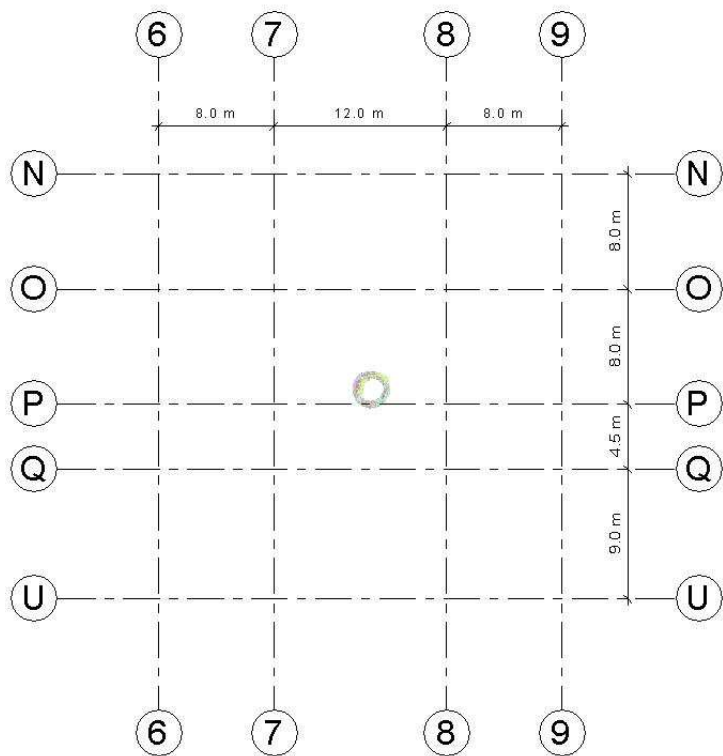


Figure 2.17: Section B-B

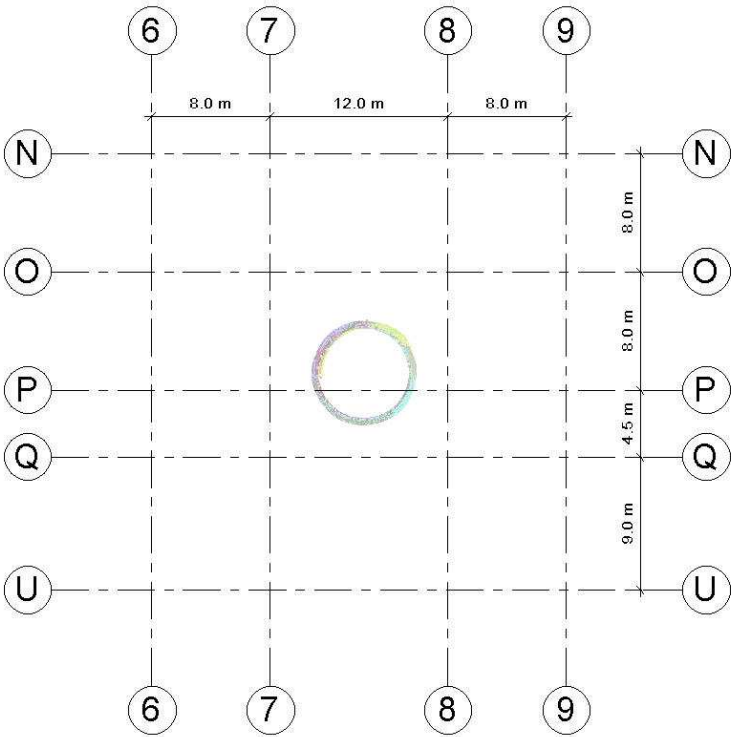


Figure 2.18: Section C-C

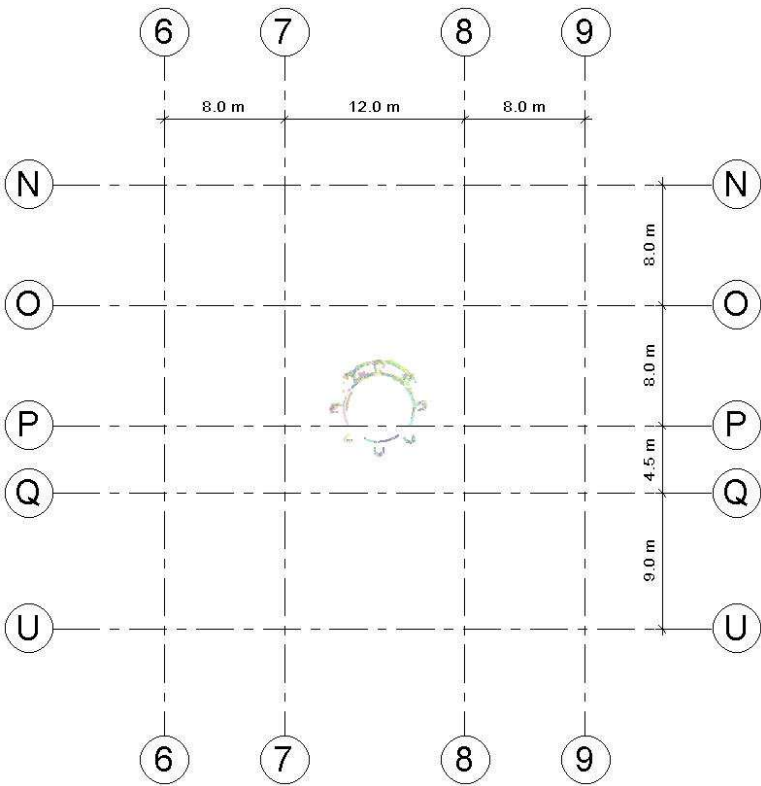


Figure 2.19: Section D-D

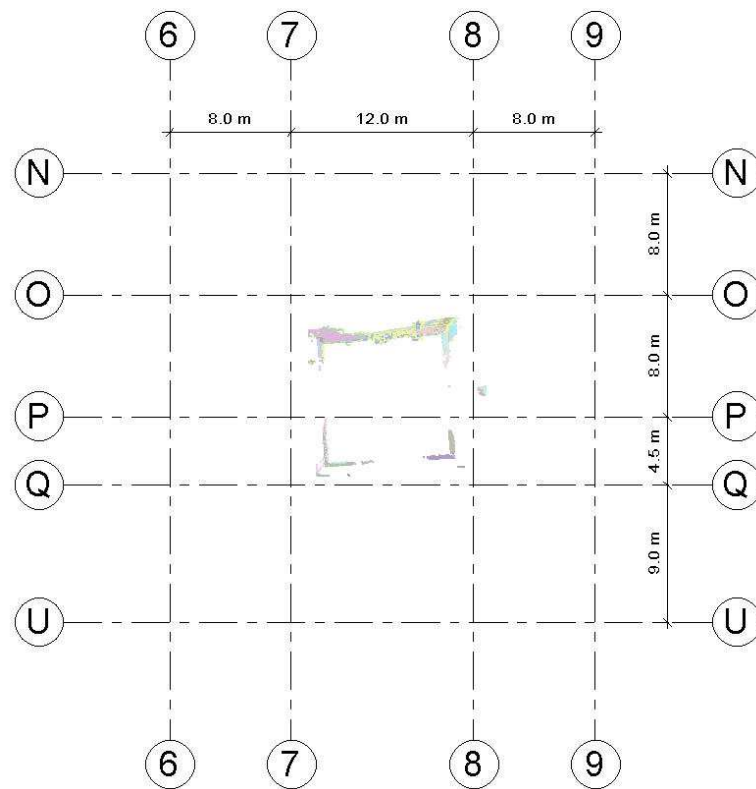


Figure 2.20: Section E-E

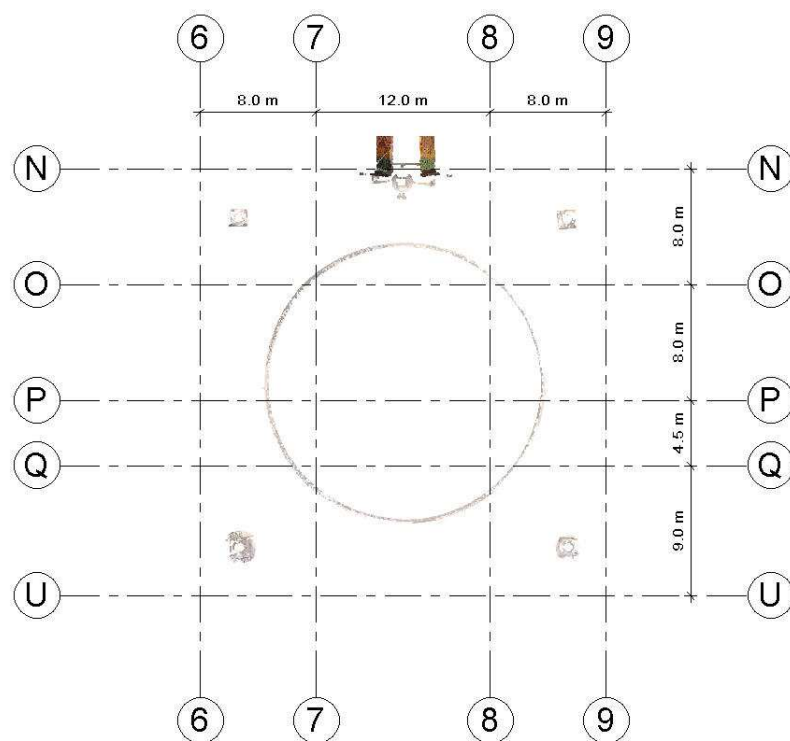


Figure 2.21: Section F-F

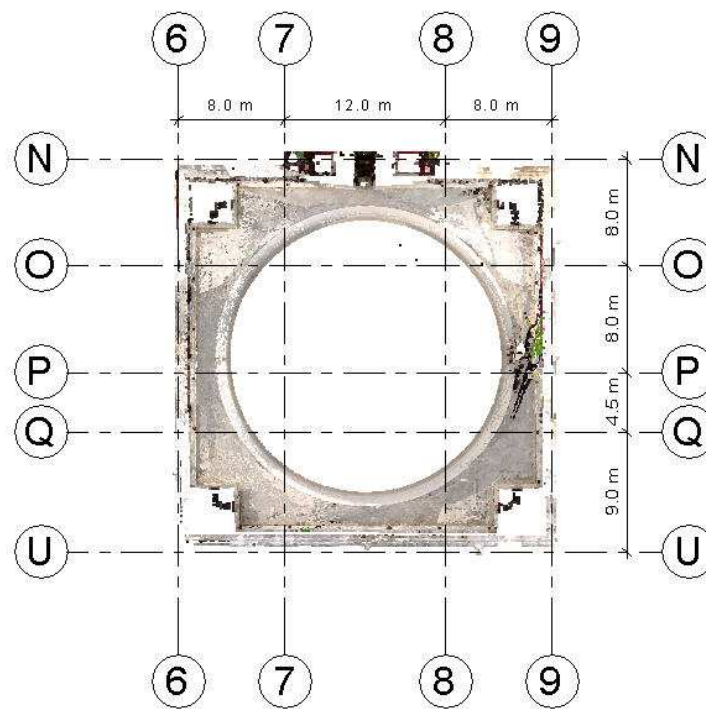


Figure 2.22: Section G-G

2.5 Data Depository

Point cloud data are stored and shared using Faro Webshare application. The data is accessible by request. Figure. 2.24 – 2.27 show samples of screenshots from Webshare application.

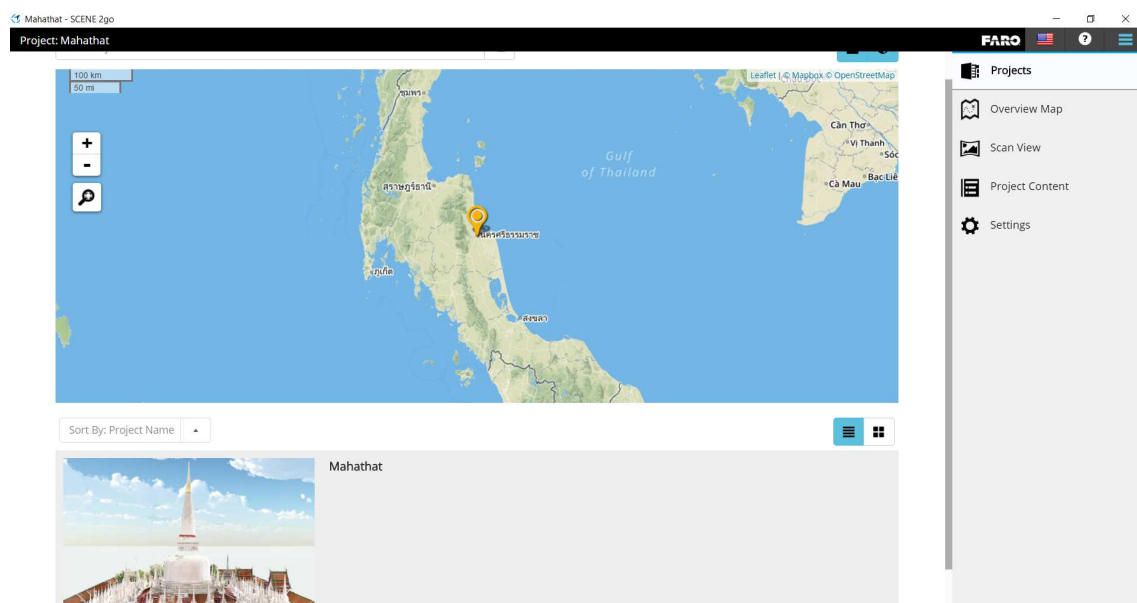


Figure 2.23: Webshare of Point Cloud Data



Figure 2.24: Webshare of Point Cloud Data



Figure 2.25: Webshare of Point Cloud Data



Figure 2.26: Webshare of Point Cloud Data

Chapter 3 3D DIGITAL DOCUMENTATION WITH IMAGE-BASED 3D MODELING

3.1 Introduction

The report presents methods to assess damages in historical buildings from surface inspection and examining the structure's geometry. The proposed method utilized the 3D model created from the image-based photogrammetry technique to estimate the tilt angle and inspect the surface damage. The 3D model was also used in Finite Element modeling to evaluate stability of the structure. The 3D model obtained from the image-based technique was also tested with the model obtained from a laser scanner (LIDAR). It was found that the 3D model from the image-based technique provides an identical result with the laser scanner result with the registration error of 0.0546. We also discover that the top of Chedi Phra Mahathat is tilted by about 1.5° . The subsequent section summarizes the methodology, data collection, and results. Top view image from UAV of the site is shown in Figure. 3.1.



Figure 3.1: Top view image from UAV of the site.

3.2 Data acquisition

In this section, an image dataset was acquired using the image-based photogrammetry technique. The point clouds from these models are used for comparison as explained in a later subsection. The image dataset is used with the software called Agisoft. There are no strict rules in taking pictures; the rule of thumb is consecutive images must have an overlap of at least 50%.

In the temple dataset, pictures were taken by a DJI Phantom 4. Generally, an automatic flight strategy called Point Of Interest flight (POI) is flying around an object and the procedure was repeated to obtain images for an entire structure. As shown in Figure. 3.2, the output from Agisoft shows a sparse point cloud and the locations of where each image was taken. It can also give a dense 3D surface model, in which the point cloud is denser and provides more realistic visualization for a building as depicted in Figure. 3.3(a).

A 3D point cloud from a laser scanner was collected for a temple as shown in Figure. 3.3(b). The 3D laser scanner data was used for comparison. The software Faro Scene was applied to register the point clouds from each location to form a single point cloud for the entire temple.

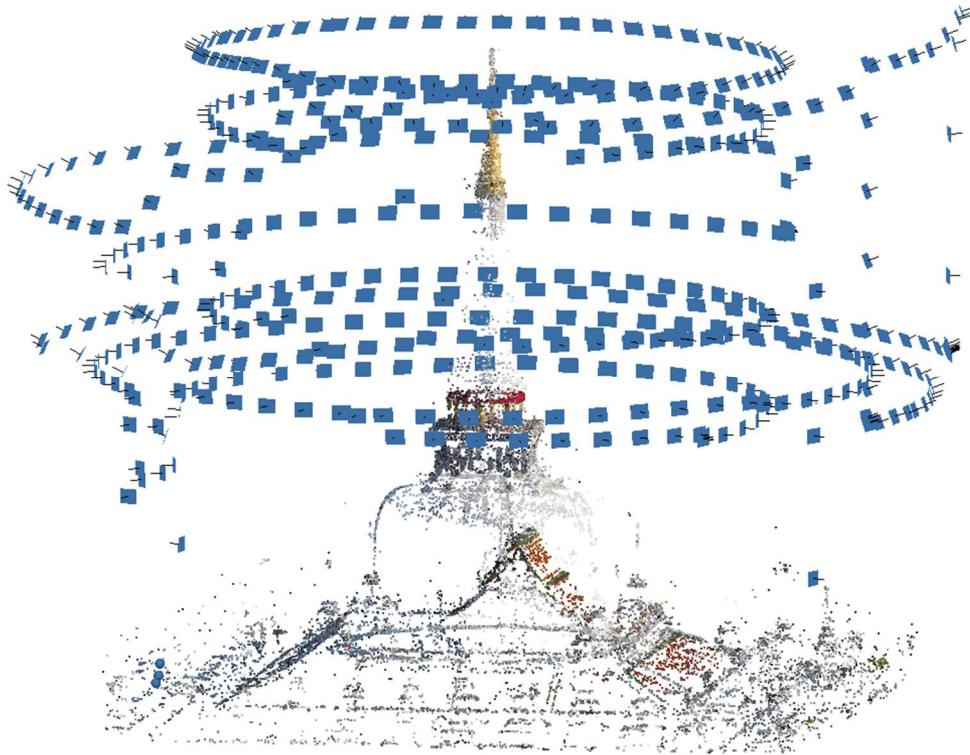


Figure 3.2: A 3D model of the temple created from Agisoft with camera positions.

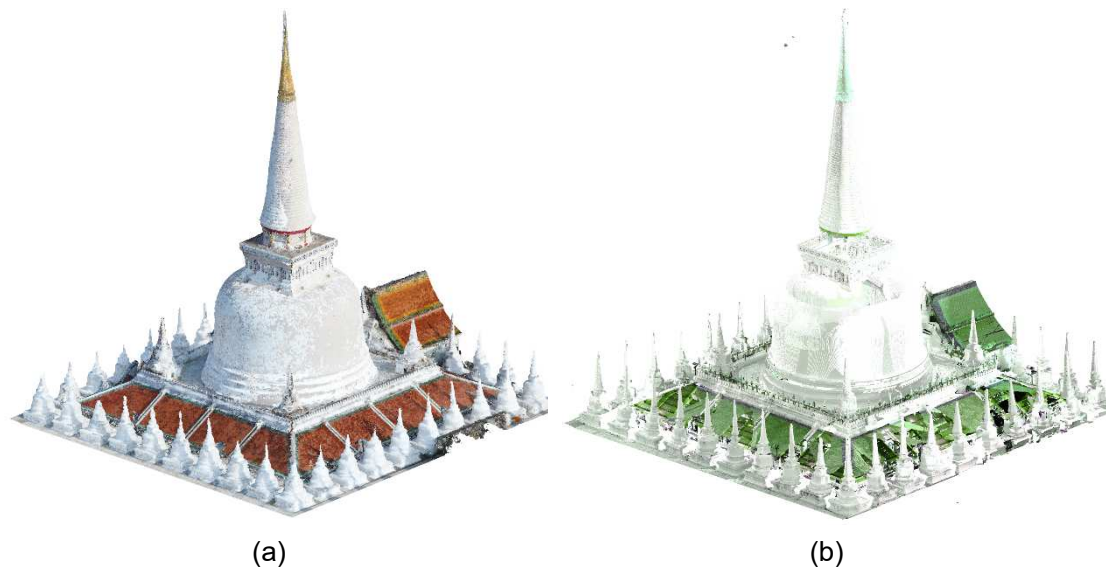


Figure 3.3: (a) 3D model from Agisoft Photoscan (b) 3D model from LIDAR

3.3 Registration

In this section, the comparison is made between models from Agisoft and LIDAR. The 3D models from Agisoft and LIDAR are registered in CloudCompare using an iterative closest point (ICP) algorithm, which is applied iteratively until RMS errors between the models are constantly taken as the final RMS error. Initial alignment between two models must be specified as a reference in order to register the models accurately. In this study, the base of the temple was used as the initial alignment. Figure. 3.4 and Tab. 3.1 shows the results from registration of the two 3D models.

It can be seen from Table 3.1 that the RMS error is 0.0546, which is very small, which can be concluded that the accuracy of the image-based technique is comparable to the laser scanning method. From Figure 4, it can be seen that the point cloud of both techniques are identical, and the results from both techniques are not different.

Table 3.1: A summary of the registration result between the Agisoft and LIDAR models.

Data	Chedi Phra Mahathat
RMS	0.0546
LIDAR #points	4,000,000
Agisoft #points	4,211,475
#points Ratio	1.052

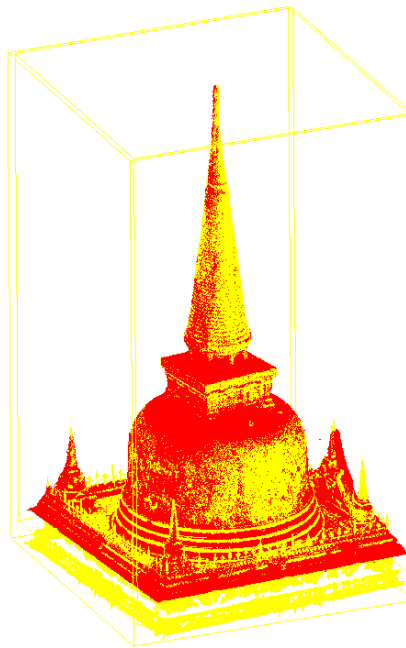


Figure 3.4: Point cloud registration between the models from Agisoft and LIDAR.

3.4 Deformation Assessment

As shown from Figure. 3.5(a), the centre line of the temple starts from the centre of the temple base, which was found by an intersection of the diagonal of the base. In this study, it was assumed that the whole temple does not tilt together as a single rigid object. The temple is split into an upper part with the height between 25 to 55 meters and a lower part with the height from 0 to 25 meters. This is based on the assumption that the top of the temple can move more than the lower part and also the point cloud is more complete at the lower part than the upper part. As shown in Figure. 3.5(b), the temple will have two tilt angles, θ_{top} for the upper section and θ_{bottom} for the lower section. From Figure 3.5(a), the tilt angle is estimated from a straight line that best fitted to the centres of each layer, in which a centre is estimated from the centre of a best-fit circle on each layer.

Figure 3.6 shows the results of tilt angles of the temple. The orange line is the LIDAR data and the blue line is the Agisoft data. Figure 3.6(a) is the cross section using South direction and Figure. 3.6(b) is the cross-section on the East direction. On Figure. 3.6, it can be seen that the tilt angles are different on upper and lower part. The tilt angles are almost identical for both datasets, which suggests that the accuracy of the model from Agisoft is similar to the model from LIDAR. The angle of inclination is between 0.5 – 1.5 degree. Figure 3.7 shows the results mapped onto the temple to show the tilt angle.

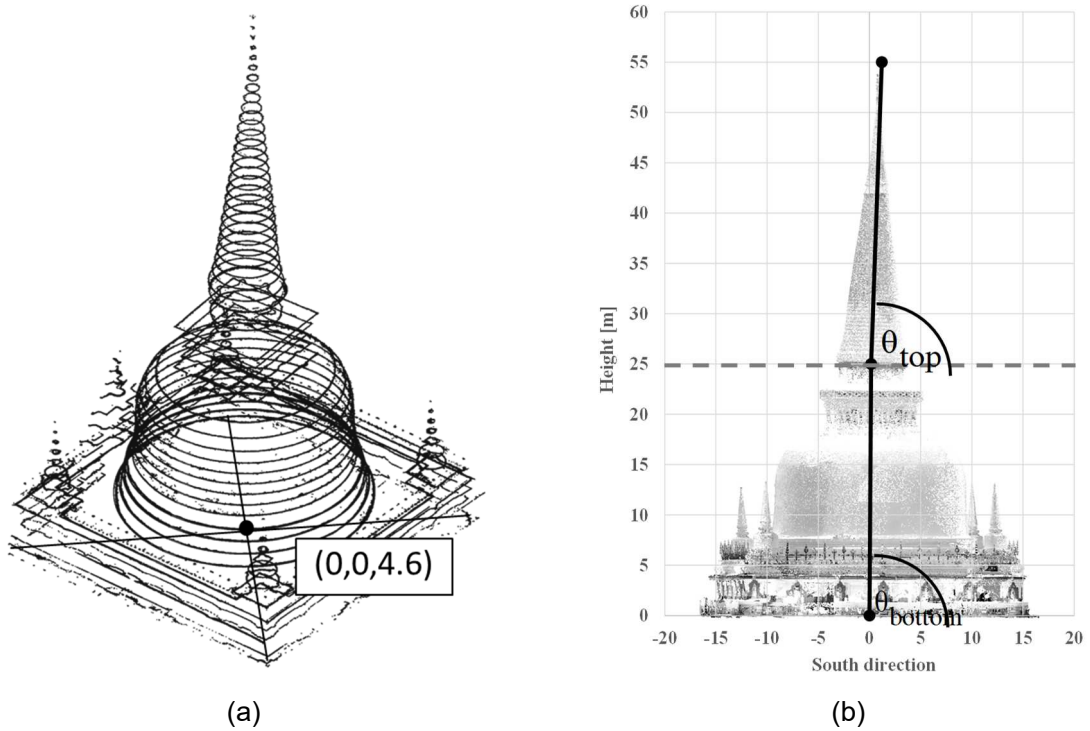


Figure 3.5: (a) Location of the center of the temple; (b) Angles of inclination of the top and bottom parts of the temple.

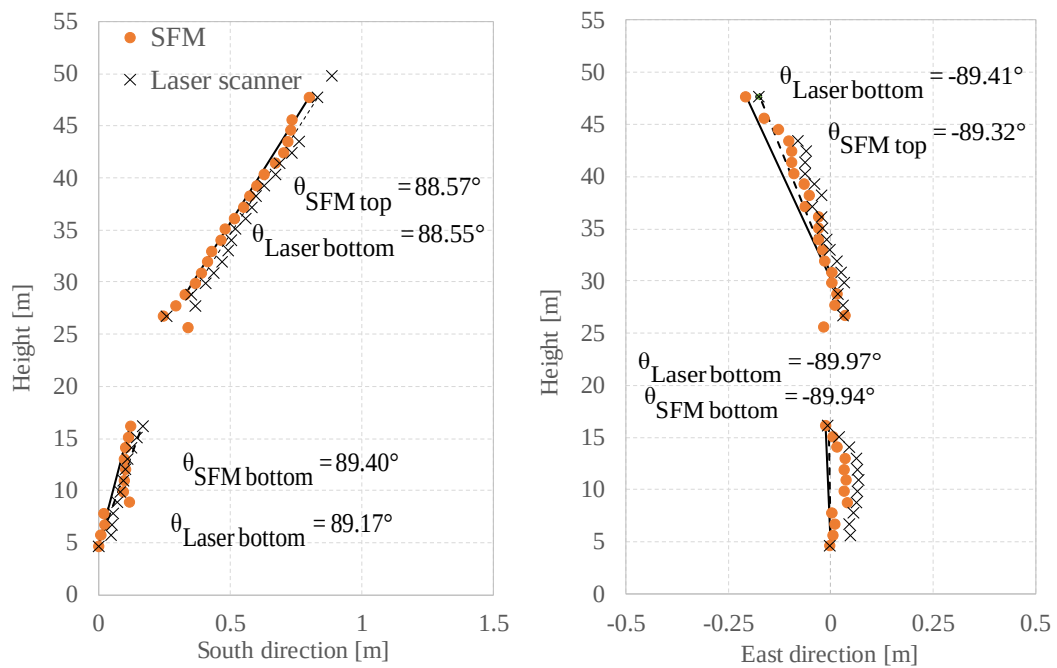


Figure 3.6: Results of inclined angles on the top part (a) South direction and (b) East direction

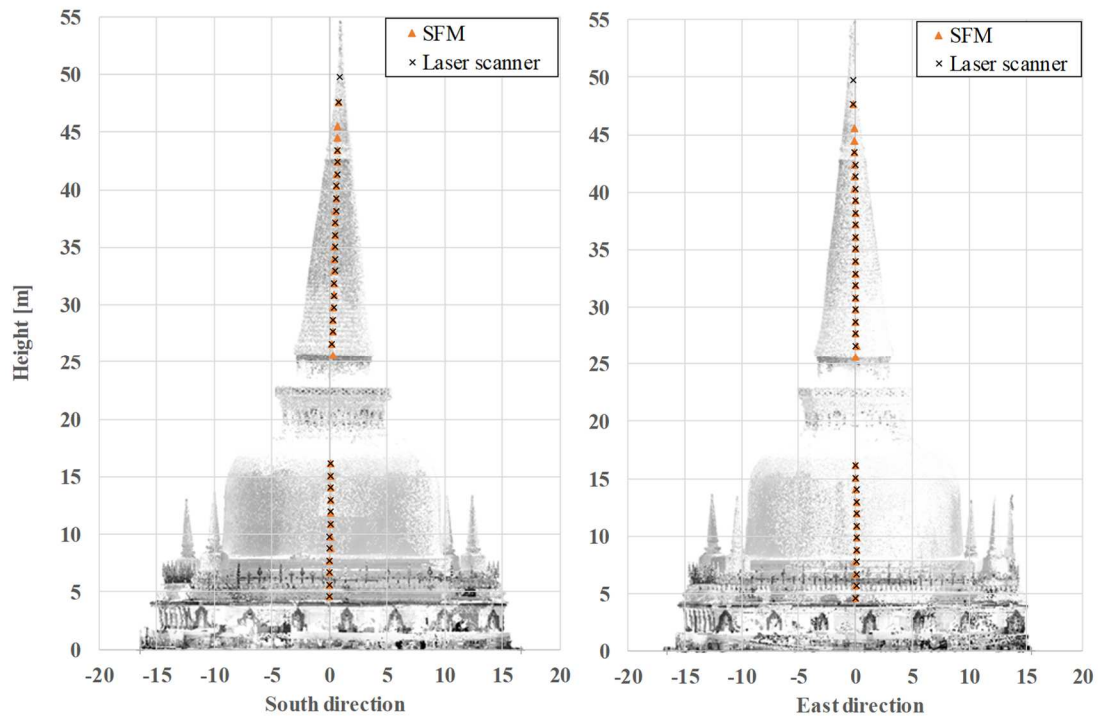


Figure 3.7: Results of inclined angles of the structure (a) South direction and
(b) East direction

3.5 SfM to solid modeling

The image-based model can be converted into a solid model manually through autodesk inventor. Figure. 3.8 shows the flight path to collect the objects around Chedi Phra Mahathat. Figure. 3.9 shows the results of the conversion from the point cloud model. Figure. 3.10, 3.11 and 3.12 shows different types of chedis in various zone in the site. It can be seen that there are various types of chedis and sizes. These solid models are used for FEM modeling. Figure. 3.13 shows the solid model of Chedi Phra Mahathat.



Figure 3.8: Flight path of chedi in zone 1

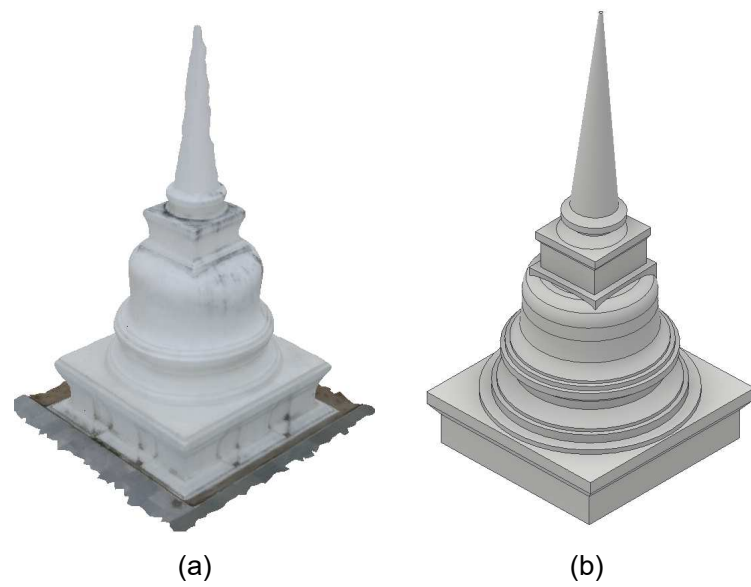


Figure 3.9: Converting 3D model, (a) mesh model (b) Solid model

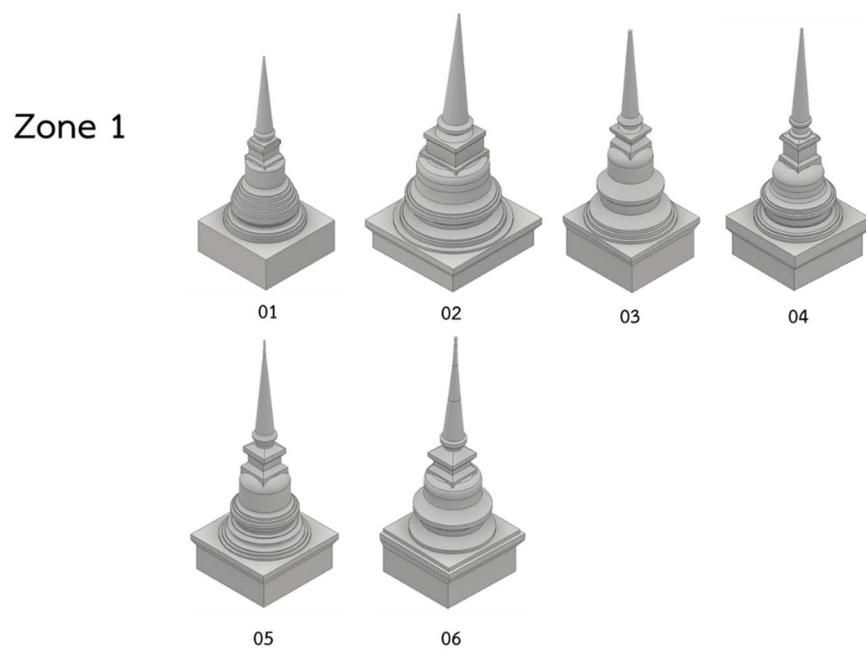


Figure 3.10: 3D model of chedi in zone 1

Zone 2

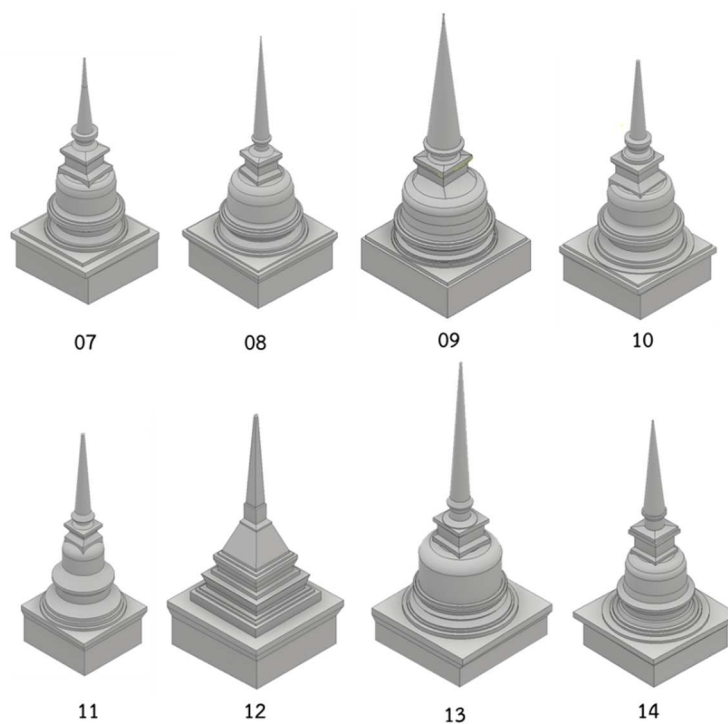


Figure 3.11: 3D model of chedi in zone 2

Zone 3

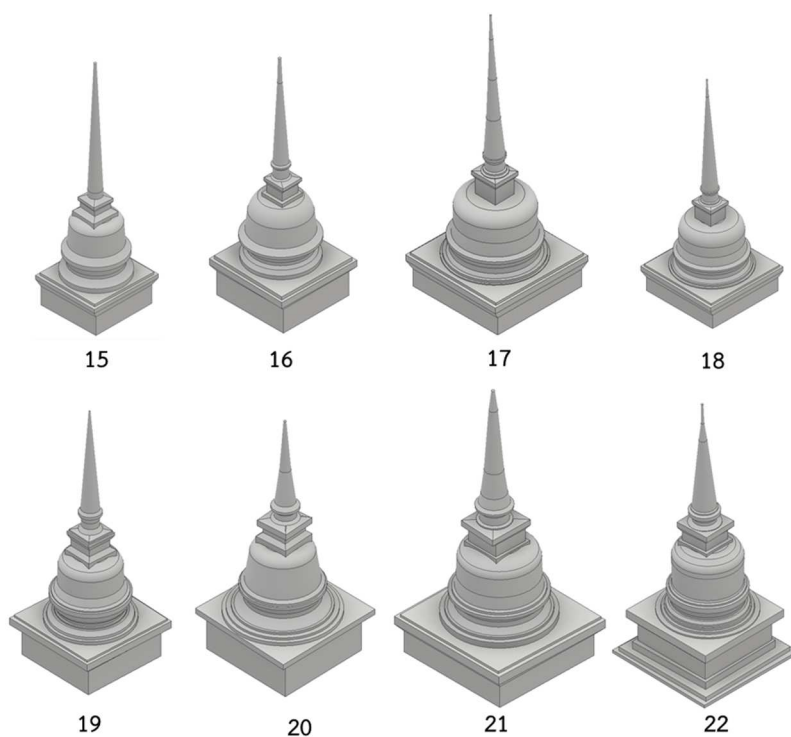


Figure 3.12: 3D model of chedi in zone 3

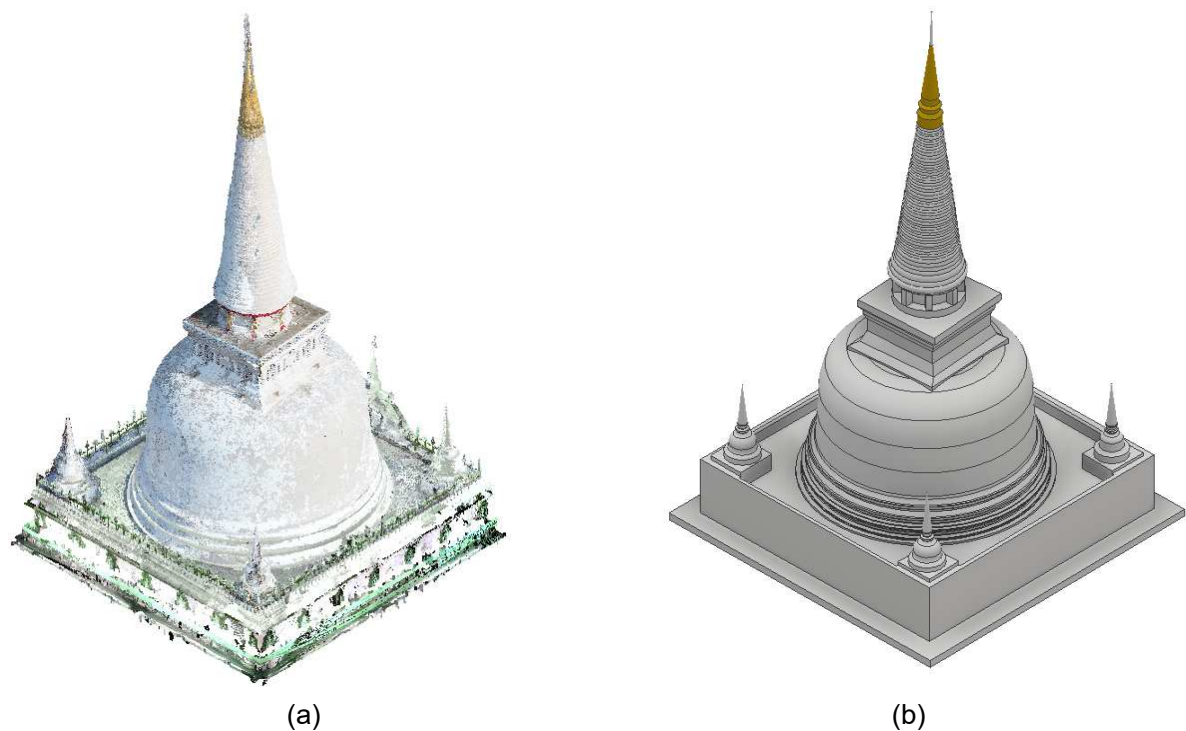


Figure 3.13: Converting 3D model, (a) point cloud (b) Solid model

3.6 A UAV based inspection for ancient structure

Figure. 3.14 and 3.15 shows examples of surface from reconstructed model from the top of Chedi Phra Mahathat. At these heights, inspection of surface damage is different. The 3D model can be used for damage inspection at high altitude. The use of image-based 3D model together with the drone allows the surface at inaccessible areas to be inspected and examined, which is complimentary to the data obtained from the laser scanner.



Figure 3.14: Surface from reconstructed model



Figure 3.15: Surface from reconstructed model

Chapter 4 IDENTIFICATION OF NATURAL FREQUENCY OF STRUCTURE BY AMBIENT VIBRATION TEST

4.1 Fundamental of the Identification Technique

Natural frequency is an important parameter of a structure as it composed of mass and elasticity information or natural properties of the structure. The natural frequency can be used to calibrate a finite element model for structural analysis. Moreover, it can be regularly monitored in order to detect any change in mass or elasticity of the structure which is beneficial for conservation and maintenance plan of an important structure.

In general, natural frequencies of structures can be calculated by numerical analysis of the structural model or identified by direct measurement of the actual structure. The numerical study yields complete results but likely to be inaccurate. In addition, effects of complex vibration characteristics as torsional eccentricity or foundation compliance may not be correctly modeled. Thus, direct measurement through dynamic test is considered to provide more reliable results. There are two forms of dynamic test on structures, forced vibration test and ambient vibration test. The forced vibration test is to measure responses of the structure under artificially controlled excitation, while the ambient vibration test is to measure responses of the structure induced by wind, traffic loads nearby and other environmental forces. The ambient vibration test has been verified to be very practical yet yields accurate results; therefore, this study applies the ambient vibration test. This test requires processing signals for identifying dynamic properties of structures. The signal processing techniques can be categorized into time domain and frequency domain technique. In the time domain technique, response-time history is employed directly in the identification of dynamic properties of structure. While in the frequency domain technique, response-time history has to be firstly converted into frequency domain by Fourier analysis. Dynamic properties are then extracted from the frequency spectrum, which is plotting of Fourier magnitude response against frequencies. Although both techniques can be employed to identify the dynamic properties of structures, the frequency domain technique gives a better physical interpretation than the time domain technique because it presents the response of structures in the form frequency spectrum. In this technique, the natural vibrations from ambient disturbances of a structure are recorded in time domain and transformed into frequency domain spectrum using Fast Fourier Transform technique. By assuming that excitations are stationary random process with a board band spectrum, the frequency spectrum can be used directly to identify natural frequencies from the frequencies corresponding to the peak values of the Fourier magnitude.

4.2 Measuring equipment

The main equipment used in the measurement is a set of velocity meters to record the response of the structure in form of velocity of motion. The velocity meters consist of 3-axis velocity sensors, a portable recorder and a notebook computer as shown in Figures. 4.1. The velocity sensors convert the vibratory velocities into analog electrical signals. The recorder further converts these analog signals into equivalent digital signals and records them in its memory. The notebook computer is used for controlling the recorder through a special data acquisition software package. The resolution of the analog to digital converter is 24 bits with maximum converting speed of 50 kHz, and maximum input voltage of 2.5 V. The sensitivity is 1 V/(cm/s) and the frequency range is from 0.5 to 20 Hz. The system can measure velocity in the scale of micro-m/s and the maximum scale is ± 2.5 cm/s.



Figure 4.1: Velocity measuring system

4.3 Ambient Vibration Tests of Wat Phra Mahathat Woramahawihan

This study conducted ambient vibration tests for 23 chedis in the temple, Chedi Phra Mahathat and 22 subsidiary chedis. The subsidiary chedis are smaller than Chedi Phra Mahathat, and having different height and shape. Figure. 4.2 shows three area of the subsidiary chedis in the measurement. The number of the studied chedis in each area is, 6, 8 and 8 for area 1, 2 and 3, respectively. The reference horizontal axes, X and Y, as defined by the vibration equipment are shown in this figure.

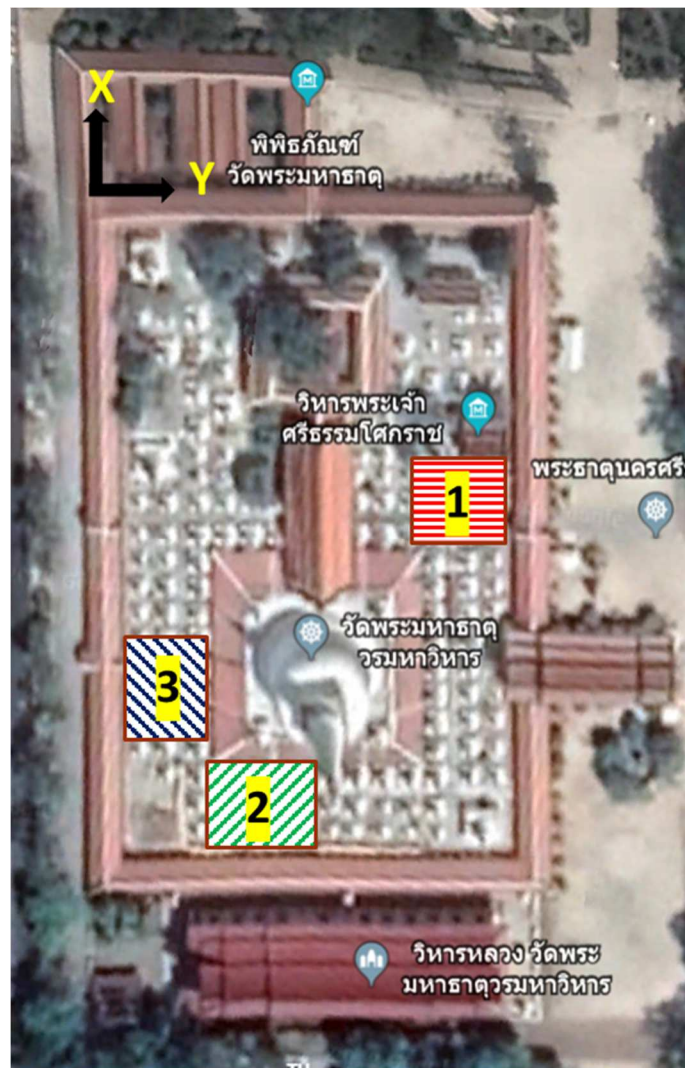


Figure 4.2: Area of the chedis selected for ambient vibration test.

4.4 Results of Chedi Phra Mahathat

The positions of the vibration sensors are shown in figure. 4.3, from elevation +0.00 to +7.86 m from ground. The locations were only at the lower part of the chedi because of restrictions to access to the higher part.

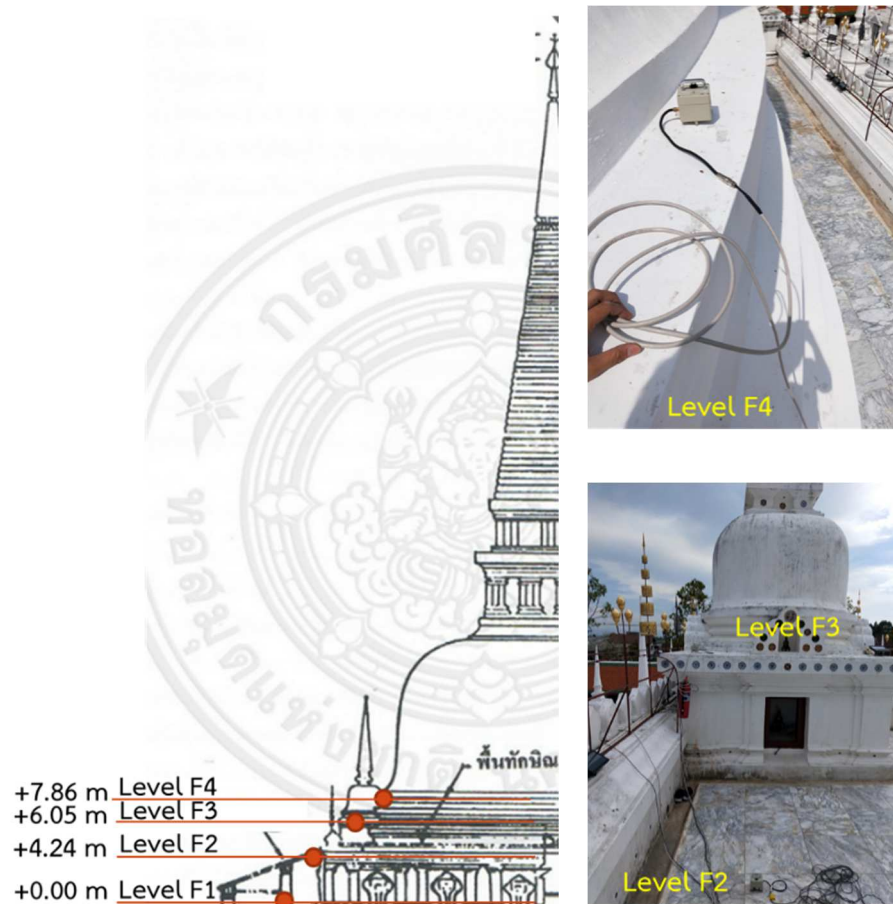


Figure 4.3: Locations of vibration sensors for ambient vibration test of Chedi Phra Mahathat

Figure 4.4 shows samples of ambient vibration records from different locations in X and Y axes. Then these data are transformed into Fourier spectra using Fast Fourier Transform technique. The plots of Fourier spectra for level F1 to F4 are shown in Figure 5.5. In this figure, the dominant peaks can be clearly identified, especially at level F4 where the location is in the highest position. This peak is about 1.96 Hz. and 1.84 Hz. In X and Y direction, respectively. These values are the identified natural frequency for the first mode of vibration of the chedi. At level F3, as shown in the second set, the dominant peaks were also observed around 10.5 Hz. In both directions. This frequency is the fundamental frequency of the satellite chedi.

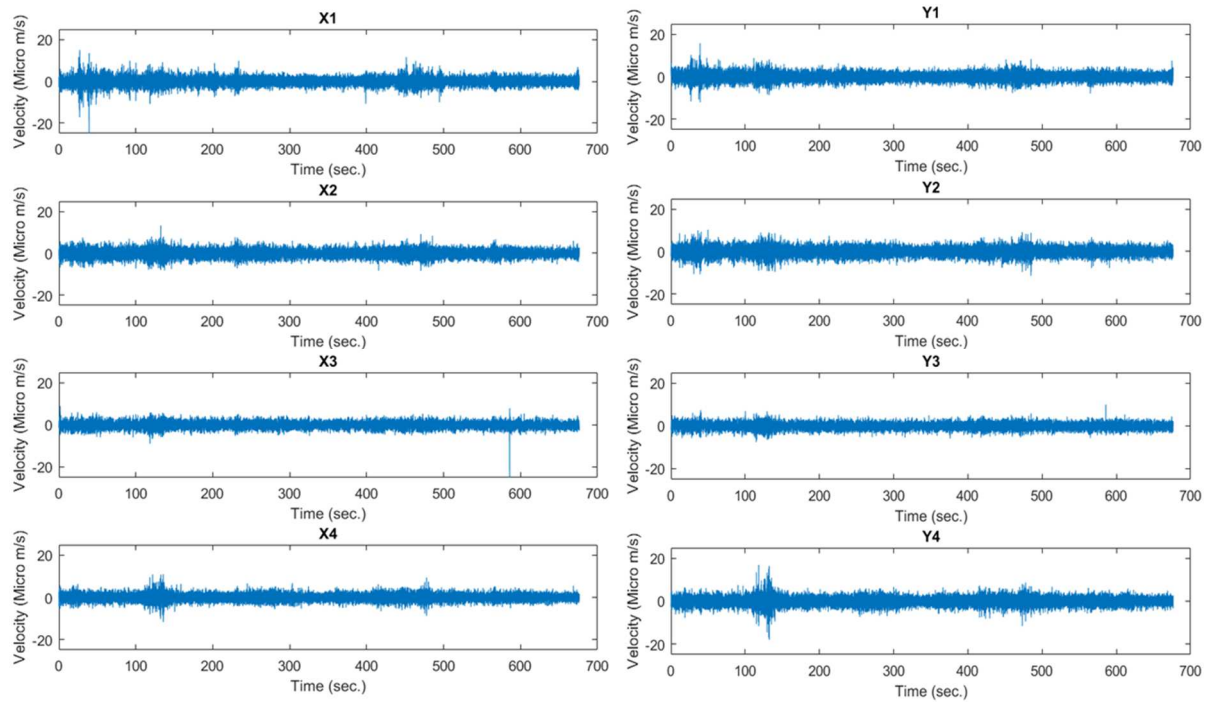


Figure 4.4: Example of ambient vibration signal

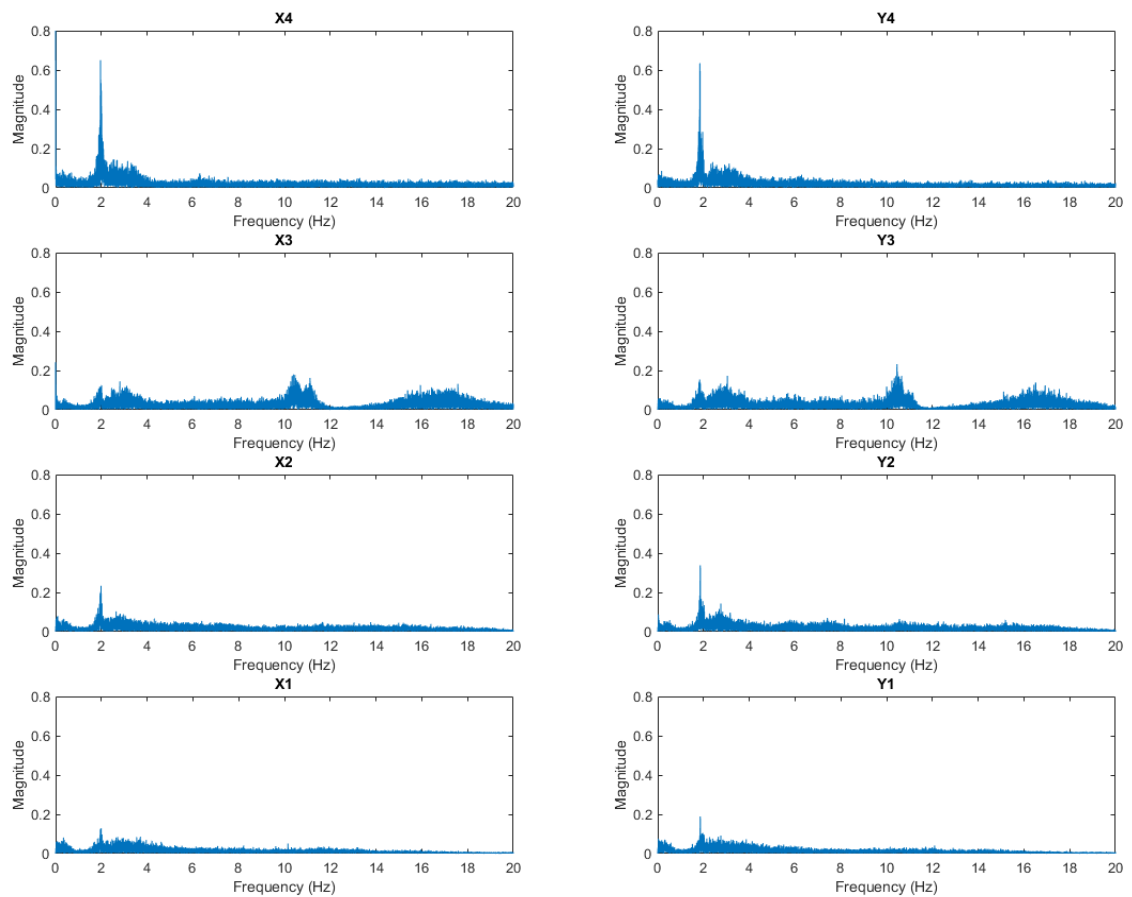


Figure 4.5: Fourier spectra plots (from top; level F4, F3, F2 and F1)

Ambient vibration tests were also conducted for 22 subsidiary chedis. Figure 4.6 shows location of the chedis in zone 1, and figure 4.7 is for the chedis in zone 2 and 3. The results of the identified natural periods of the fundamental mode and height of each chedi are presented in table 4.1. The plot of the natural periods and height is shown in figure 4.8. The natural periods increase with height of the chedi.

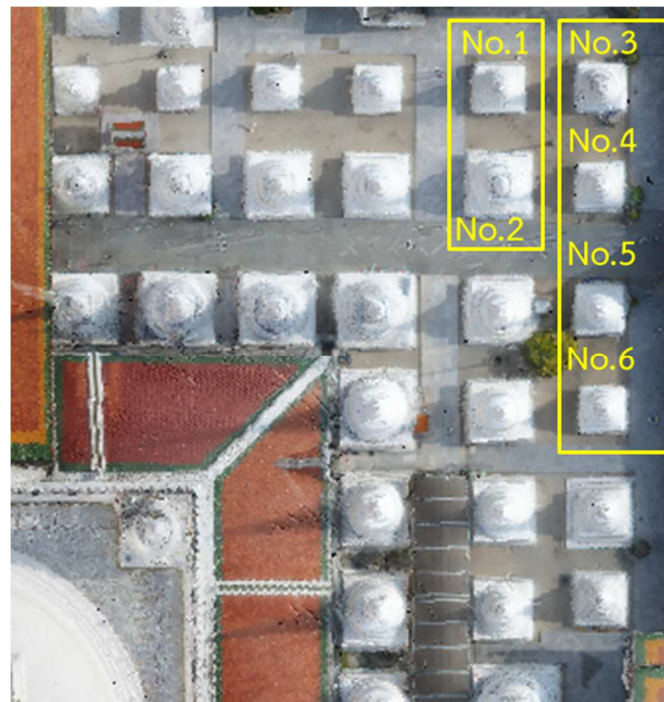


Figure 4.6: The chedis for measurement in zone 1

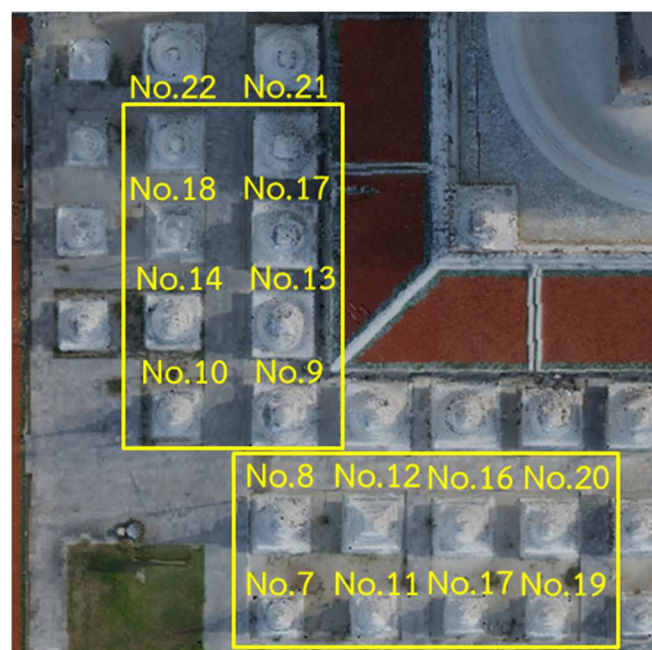


Figure 4.7: The chedis for measurement in zone 2 and 3

Table 4.1: Natural period of the fundamental mode of the investigated chedis

Name	H (m)	period [sec.]	
		T1x	T1y
Chedi 01	5.82	0.08	0.08
Chedi 02	7.12	0.09	0.09
Chedi 03	6.21	0.09	0.09
Chedi 04	6.09	0.08	0.08
Chedi 05	6.47	0.09	0.10
Chedi 06	6.32	0.09	0.09
Chedi 07	5.39	0.09	0.08
Chedi 08	7.63	0.11	0.11
Chedi 09	9.39	0.17	0.16
Chedi 10	6.83	0.10	0.11
Chedi 11	5.92	0.10	0.09
Chedi 12	6.54	0.07	0.08
Chedi 13	9.41	0.17	0.16
Chedi 14	7.37	0.10	0.11
Chedi 15	7.45	0.13	0.14
Chedi 16	8.04	0.12	0.12
Chedi 17	9.69	0.17	0.17
Chedi 18	6.97	0.09	0.09
Chedi 19	6.11	0.09	0.09
Chedi 20	6.34	0.09	0.09
Chedi 21	7.73	0.11	0.12
Chedi 22	7.50	0.10	0.10

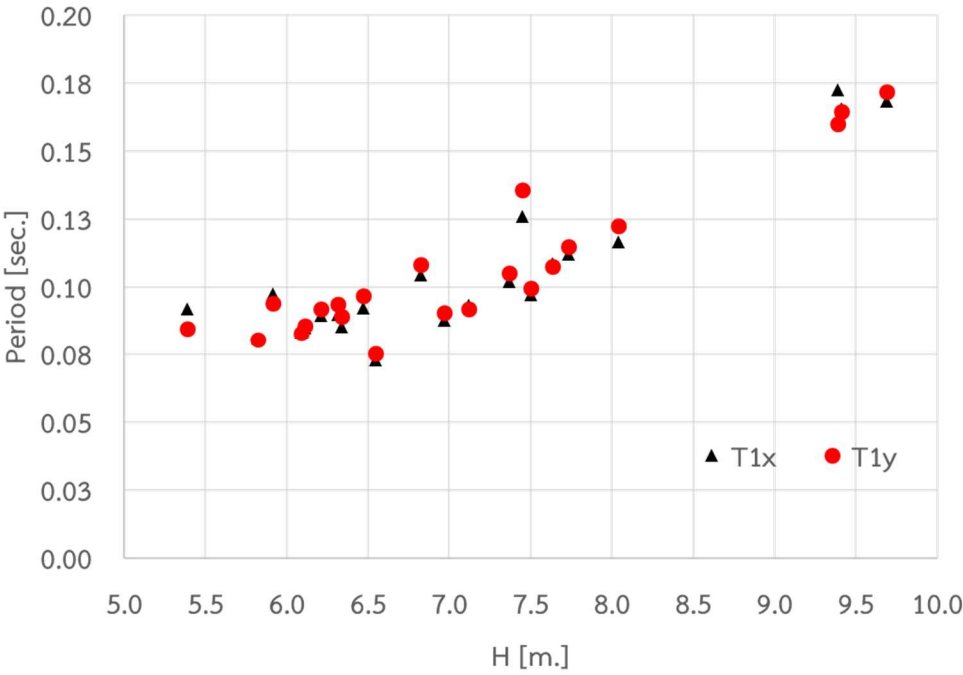


Figure 4.8: Relations of the natural period and height of the subsidiary chedis

Chapter 5 FINITE ELEMENT MODELING OF CHEDI PHRA MAHATHAT

This section presents the finite element modeling of Chedi Phra Mahathat which is the Chedi Phra Mahathat of Wat Mahathat Woramahawihan, Nakhon Si Thammarat, Thailand. The section is divided into six parts. The first part describes the geometric model of Chedi Phra Mahathat and finite element discretization of the model. The second part describes the properties of material necessary for the analysis under the linear-elastic assumption, and the use of vibration data to estimate the modulus of elasticity. The third part presents the stress analysis of the structure under its own weight together with the safety evaluation of the structure. The fourth part presents the modal dynamic properties of the structure which are important for dynamic analysis. The fifth part is the study on the effect of tilting on the safety of the structure. The last part is the summary of the section. The ANSYS Workbench R19 / Academic Version [1] was used throughout the finite element modeling process.

5.1 Geometric modeling and finite element discretization

Geometric information from the Structure from Motion technique and 3-D laser scan were used to model part of the structure that is above ground level. The base is rectangular with a size of $27.7 \times 28.0 \text{ m}^2$. Tak Sin Floor is at +4.28 m. above ground level and the structure has a total height (above ground) of 54.73 m. Bottom level of the base is at -2.90 m. below ground level as reported by [2] and [3]. Satellite pagodas and Wihan Tab Kaset are excluded from the model since their effects on the analysis are limited due to their relatively small sizes as compared to the. The structure is assumed to be isolated from Wihan Song Ma on the North. Figures 5.1 and 5.2 show plan view and side view of the model respectively.

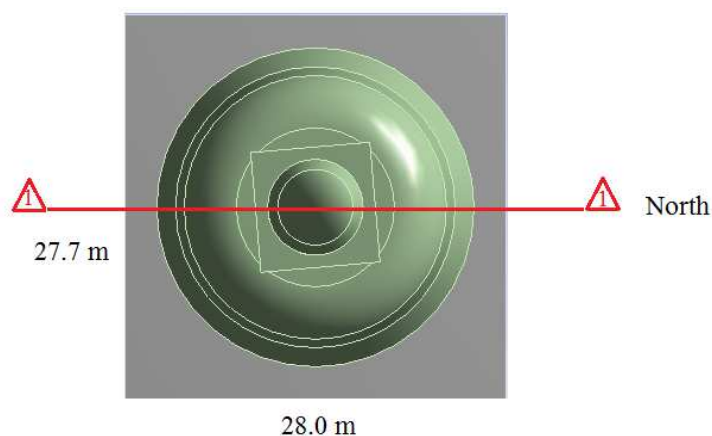


Figure 5.1: Plan view of Chedi Phra Mahathat model

The global x-axis and y-axis of the model are pointing to the North and the West respectively. Figure 5.3 shows the three-dimensional view of the model and the section 1-1 across the model. The model has a total volume of $10,075 \text{ m}^3$ and the center of mass is at $+3.35 \text{ m}$. above ground. The model is discretized into 4,157 quadratic elements with 9,834 nodes as shown in Figure 5.4. All nodes on surface at base level is fixed.

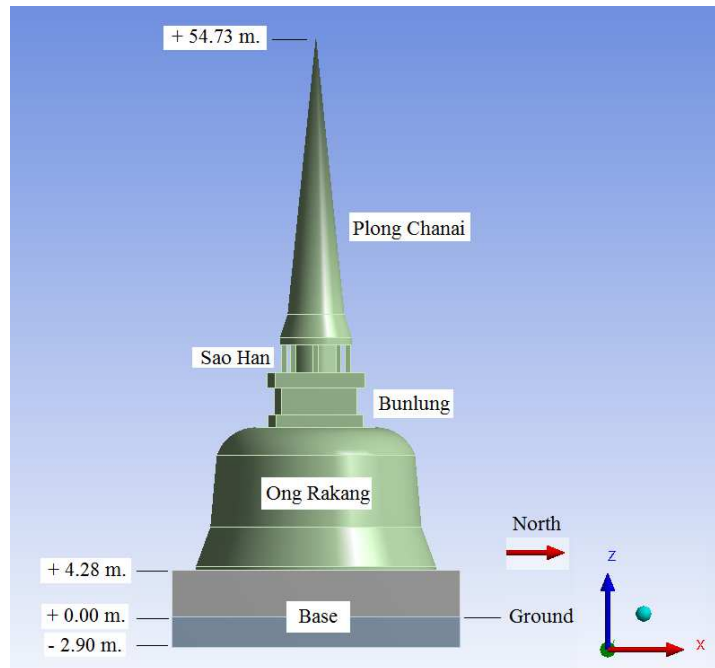


Figure 5.2: Side view of Chedi Phra Mahathat model

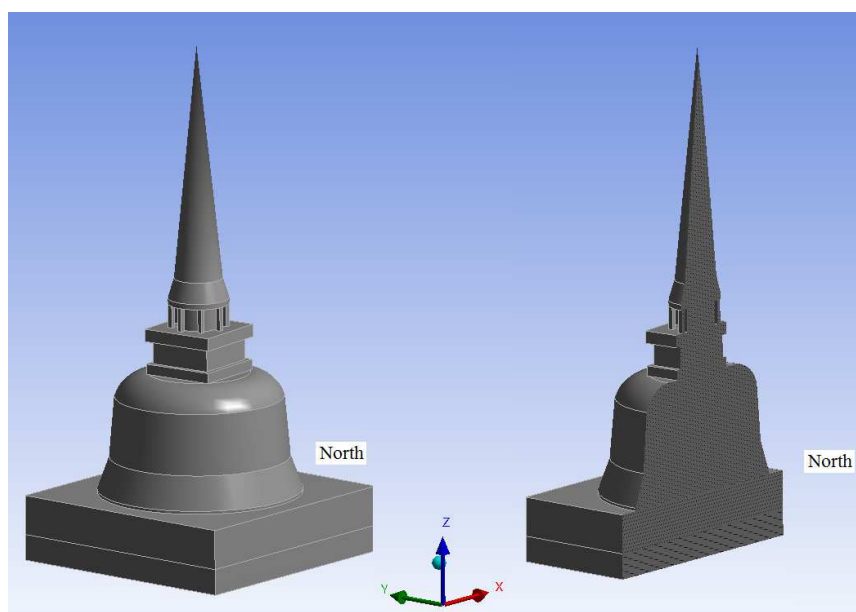


Figure 5.3: Three-dimensional view (left) and section 1-1 (right)

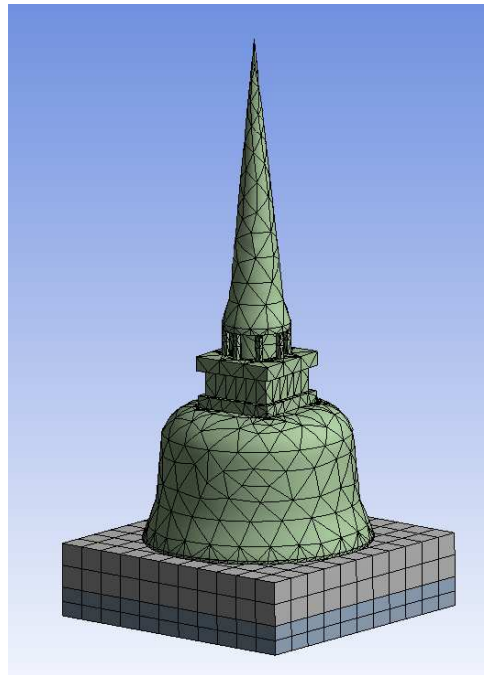


Figure 5.4: Finite element mesh

5.2 Material properties

It is assumed that the material is linear-elastic isotropic and homogeneous material. The important properties for this type of materials are mass density (ρ), Poisson's ratio (ν), and modulus of elasticity (E). It is reported in [2] that the dry density of samples taken directly from Chedi Phra Mahathat is between $1,710 \text{ kg/m}^3$ to $1,860 \text{ kg/m}^3$, and with the average moisture content of 6% the average density is about $1,850 \text{ kg/m}^3$. Accordingly, with no further information, the mass density in this study is assumed to be $1,850 \text{ kg/m}^3$. With the volume of $10,075 \text{ m}^3$, the total weight of the structure is approximated to be 182.84 MN. Poisson's ratio of the material plays a limited role in the analysis and a typical value of 0.2 is assumed.

Stiffness of the structure depends directly on the modulus of elasticity of the material. With the assumption that the material is isotropic and homogeneous material and can be represented by a single modulus, the modulus in this study is defined as the macroscopic modulus. Since stiffness plays an important role on how the structure responds dynamically, it is possible to use dynamic properties of structure to estimate for the appropriate value of the modulus.

Analysis of vibration data of Chedi Phra Mahathat indicates that the natural frequency of Chedi Phra Mahathat is about 1.90 Hz. With the mass density of $1,850 \text{ kg/m}^3$ and Poisson's ratio of 0.2,

it was found that the natural frequency of 1.90 Hz is obtained when the elastic modulus of the finite element model is about 2,200 MPa.

5.3 Stress analysis under self-weight

Brick and mortar are the primary materials of Chedi Phra Mahathat. These materials are brittle in which their compressive strengths (σ_c) are much higher than their tensile strengths (σ_t). Failure of brittle material can be described by Mohr-Coulomb failure criteria which involve the maximum principal (σ_1) and the minimum principal stress (σ_3). In this study, tensile stress is considered positive and compressive stress is considered negative.

Figure 5.5 shows the distribution of the maximum principal stress on the model. It can be seen that σ_1 of most regions are in compressive stresses except the bottom of Plong Chanai, the upper part of Ong Rakang, and the outer part of Tak Sin Floor. Maximum compressive stress of σ_1 is 81 kPa at bottom of base, and maximum tensile stress of σ_1 is 89 kPa at small regions on top of the base. Figure 5.6 shows the distribution of the minimum principal stress on the model. It can be seen that σ_3 of the whole model are in compressive stresses. Maximum compressive stress of σ_1 is 484 kPa at Sao Han region.

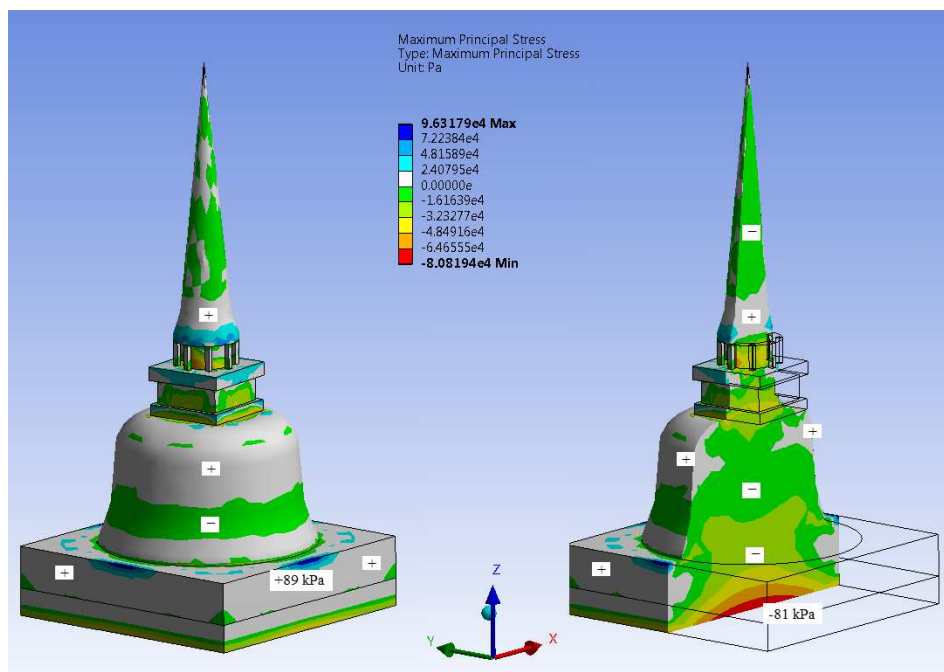


Figure 5.5: Maximum principal stress

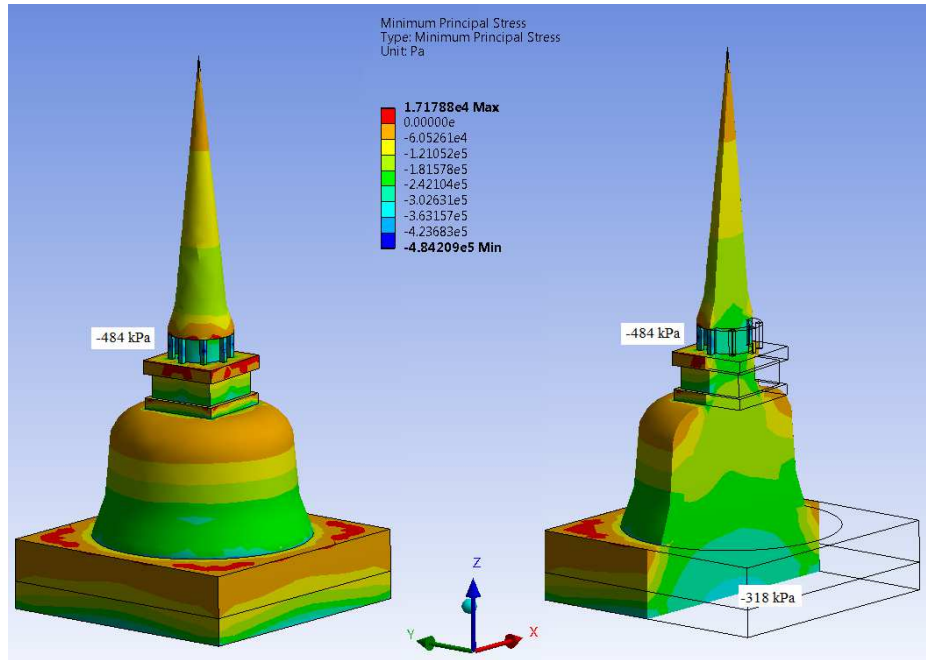


Figure 5.6: Minimum principal stress

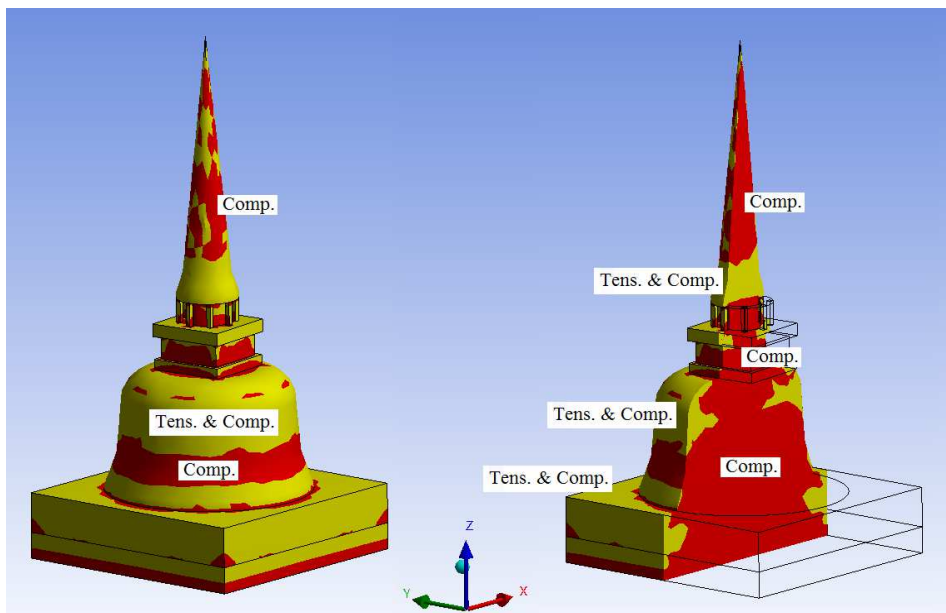


Figure 5.7: Distribution of compressive

From the distributions of maximum and minimum principal stresses, the compressive zone where both σ_1 and σ_3 are compressive stresses is shown by red color in Figure 5.7. According to the Mohr-Coulomb criteria, failure of this compressive zone happens when the demand/strength ratio $|\sigma_3|/\sigma_c$ is greater than 1.00. Also shown by yellow color in Figure 5.7 is the tensile & compressive zone where σ_1 is tensile stress while σ_3 is compressive stress. Failure of this tensile & compressive zone happens when the demand/strength ratio $\sigma_1/\sigma_t + |\sigma_3|/\sigma_c$ is greater than 1.00.

Safety factor is a simple index to evaluate safety of a state of stress and is defined as the inverse of the demand/strength ratio. State of stress is considered as a safe state when the safety factor is greater than 1.00.

The precise values of σ_C and σ_T for materials of Chedi Phra Mahathat can be known by laboratory testing of samples taken from the site. Unfortunately the information is not available in this study and the strengths must be estimated by referring to past researches. Samples of ancient bricks from eight temples in Ayutthaya, ranging from 50 to 430 years old, were tested to study their mechanical properties by [4]. It was found that the compressive strengths are between 3,700 to 5,700 kPa. Recently ancient bricks and mortars from Ayutthaya area were tested by [5], it was reported that the compressive strengths of bricks are between 2,200 to 6,702 kPa, and the compressive strengths of mortars are between 1,880 to 2,710 kPa.

In this study, the compressive strength is taken as 2000 kPa which is a rather conservative value. The tensile strength is assumed to be 200 kPa which is 10% of the compressive strength. Figure 5.8 shows the distribution of safety factor on the model. The minimum safety factor on the structure is 2.07 at the Sao Han region and the base of Plong Chanai. It is concluded that the structure is in a safe state under its own weight.

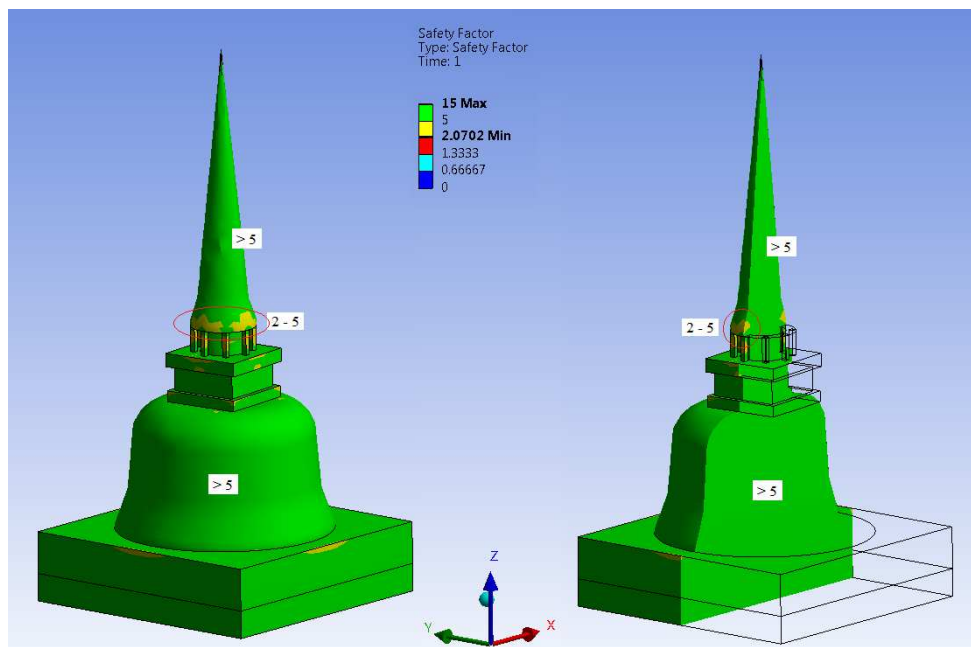


Figure 5.8: Distribution of safety factor under self-weight

Figures 5.9 and 5.10 show the distribution of bearing stress beneath the base of the structure. The average bearing stress is 247 kPa. The maximum bearing stress is found to be 320 kPa at

the center. Along the N-S direction, the stress decreases gradually to 220 kPa as it moves away from the center and increases rapidly to 320 kPa again at the edge. It is reported in [2] that the estimated bearing capacity of the soil at site is around 75 to 121 ton/m² or 736 to 1,187 kPa. If the capacity of 736 kPa is used as a reference, the safety factor for the bearing load is in the range of 2.3 to 3.7.

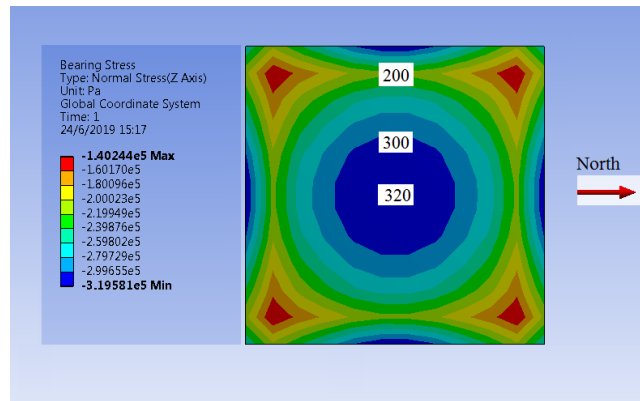


Figure 5.9: Distribution of bearing stress beneath the base

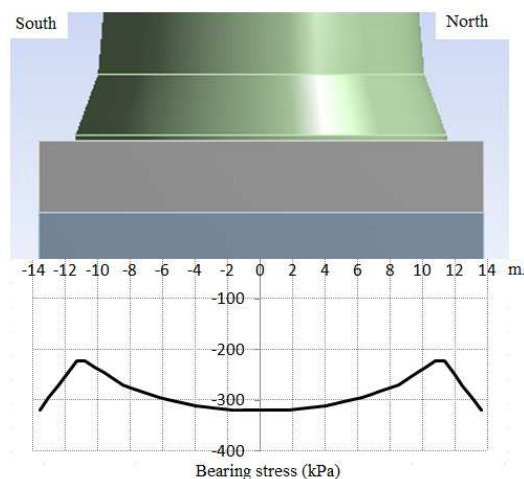


Figure 5.10: Distribution of bearing stress along N-S

5.4 Modal dynamic properties

Modal dynamic properties of the structure are the important information for analysis of structure subjected to wind load and seismic load. With the finite element model and the specified material properties, modal dynamic properties of the model can be calculated. Several modes of vibration must be included in the modal analysis in order to obtain accurate results. It is a general rule in practice that the cumulative effective masses of all modes included in the analysis should be about 85% to 90% of the total mass. According to this requirement, it is found that the first 30 modes of vibration are required in dynamic analysis of Chedi Phra Mahathat.

Table 5.1 shows the frequencies and effective masses of the first 30 modes of the structure, and Figure 5.11 shows the first three vibration mode shapes in x-direction. It can be observed from Table 5.1 that frequencies of the 1st and 2nd mode of the finite element model (1.888 Hz. and 1.889 Hz.) are very close to the value from measurement of 1.90 Hz.

Table 5.1: Modal frequency and effective mass

MODE	FREQUENCY	X-DIR	RATIO%	EFFECTIVE MASS			
				Y-DIR	RATIO%	Z-DIR	RATIO%
1	1.888	824.9	0.00	0.4436E+06	2.38	0.3648E-02	0.00
2	1.889	0.4436E+06	2.38	818.8	0.00	0.1401E-02	0.00
3	4.579	0.7781E+06	4.18	0.1019E+06	0.55	0.8612E-04	0.00
4	4.581	0.1020E+06	0.55	0.7795E+06	4.18	0.5634E-02	0.00
5	6.981	0.5252E+07	28.18	0.1458E+05	0.08	0.5342E-01	0.00
6	6.986	0.1487E+05	0.08	0.5278E+07	28.32	0.1526E-02	0.00
7	9.198	0.1922E+07	10.31	0.3711E+05	0.20	0.2395	0.00
8	9.222	0.3716E+05	0.20	0.1902E+07	10.20	0.6428E-08	0.00
9	11.01	0.1526	0.00	0.8066	0.00	1.807	0.00
10	11.72	4.818	0.00	0.7513E-01	0.00	0.5438E+07	29.18
11	12.72	0.5057	0.00	1.760	0.00	0.2274	0.00
12	13.23	0.1296E+07	6.95	0.1103E+06	0.59	28.39	0.00
13	13.28	0.1131E+06	0.61	0.1320E+07	7.08	5.819	0.00
14	16.52	0.2466E+07	13.23	0.3081E+06	1.65	41.33	0.00
15	16.58	0.3118E+06	1.67	0.2493E+07	13.37	24.38	0.00
16	17.05	48.71	0.00	4.520	0.00	0.5434E+07	29.15
17	19.85	0.1161E+07	6.23	0.1543E+06	0.83	1.264	0.00
18	20.00	0.1515E+06	0.81	0.1131E+07	6.07	1.355	0.00
19	21.87	13.72	0.00	0.4196	0.00	0.1420E-01	0.00
20	23.65	39.70	0.00	0.1049	0.00	0.1544E-03	0.00
21	24.23	0.4803E+06	2.58	0.7901E+05	0.42	0.3059E-01	0.00
22	24.48	0.8065E+05	0.43	0.4881E+06	2.62	0.6482	0.00
23	26.13	0.9659E-01	0.00	0.3643E-02	0.00	0.2437E-01	0.00
24	26.61	3.828	0.00	0.9228E-01	0.00	0.2025	0.00
25	26.82	0.7902E+06	4.24	0.1417E+06	0.76	4.116	0.00
26	26.84	0.1355E+06	0.73	0.7545E+06	4.05	2.427	0.00
27	27.14	7.013	0.00	8.477	0.00	83.91	0.00
28	27.28	2.621	0.00	0.1043	0.00	0.7397E+06	3.97
29	28.85	0.1528E+06	0.82	0.3092E+05	0.17	0.4976	0.00
30	28.87	0.3018E+05	0.16	0.1490E+06	0.80	0.1868	0.00
sum		0.1572E+08	84.35	0.1572E+08	84.33	0.1161E+08	62.31

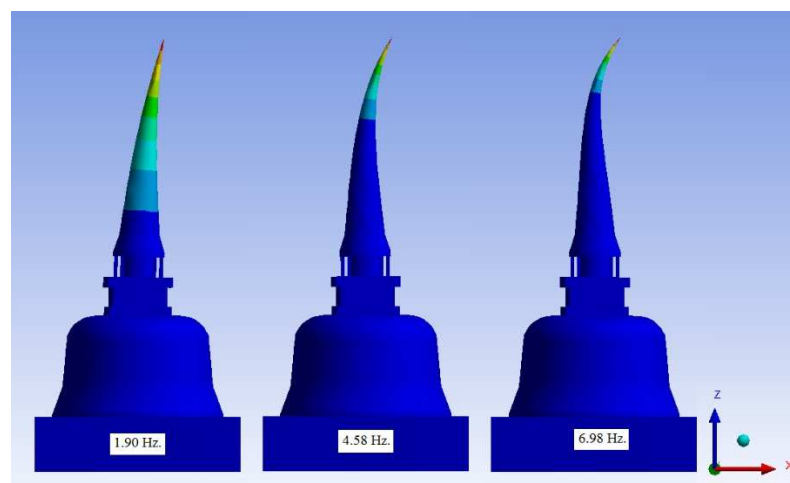


Figure 5.11: First three vibration modes in x-dir

5.5 Stress analysis under self-weight of tilting model

Results from Structure from Motion Technique and 3-D laser scan indicate that the tip of Chedi Phra Mahathat is shifted slightly from the center to the S-E direction. This fact causes a concern

to the safety of the structure, especially if the shifting continues. Assuming that shifting of the tip is caused by tilting of the structure toward the S-E direction as shown in Figure 5.12, stress analysis of the structure under self-weight with different tilting angles (θ) can be investigated. This is achieved by decomposing the gravitational force into x, y, and z axes of the analysis domain. For a tilting angle θ , the components of gravitational acceleration (g) in x, y, and z directions are $g \cdot \sin\theta \cdot \cos 45^\circ$, $g \cdot \sin\theta \cdot \sin 45^\circ$, and $g \cdot \cos\theta$ respectively.

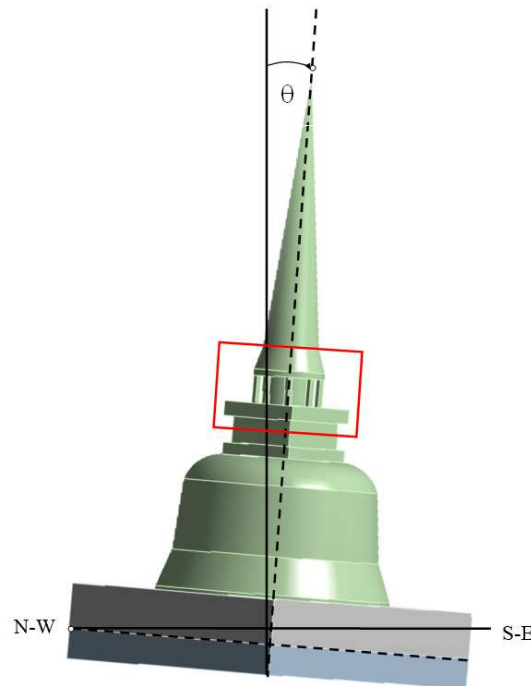


Figure 5.12: Tilting of the model

It was found that tilting has significant effect on the safety of Sao Han and the zone between Plong Chanai and Bunlung as shown by the red frame in Figure 5.12. Figure 5.13 shows the safety factor of this region for tilting angle of 0° to 8° . It was found that as the tilting angle increases, the compressive stress of Sao Han at S-E corner increases but the compressive stress of Sao Han at N-W corner decreases. Safety factor of Sao Han at S-E corner is reduced from being greater than 3 for 0° to be less than 2 for $\theta > 4^\circ$. However the safety factor is still greater than 1 even for a tilting angle of 8° . Tensile stress begins to develop in N-W Sao Han at a tilting angle of 5° . The presence of tensile stress greatly reduces the safety factor of Sao Han at N-W corner and it becomes lower than 1 when the tilting angle is greater than 7° .

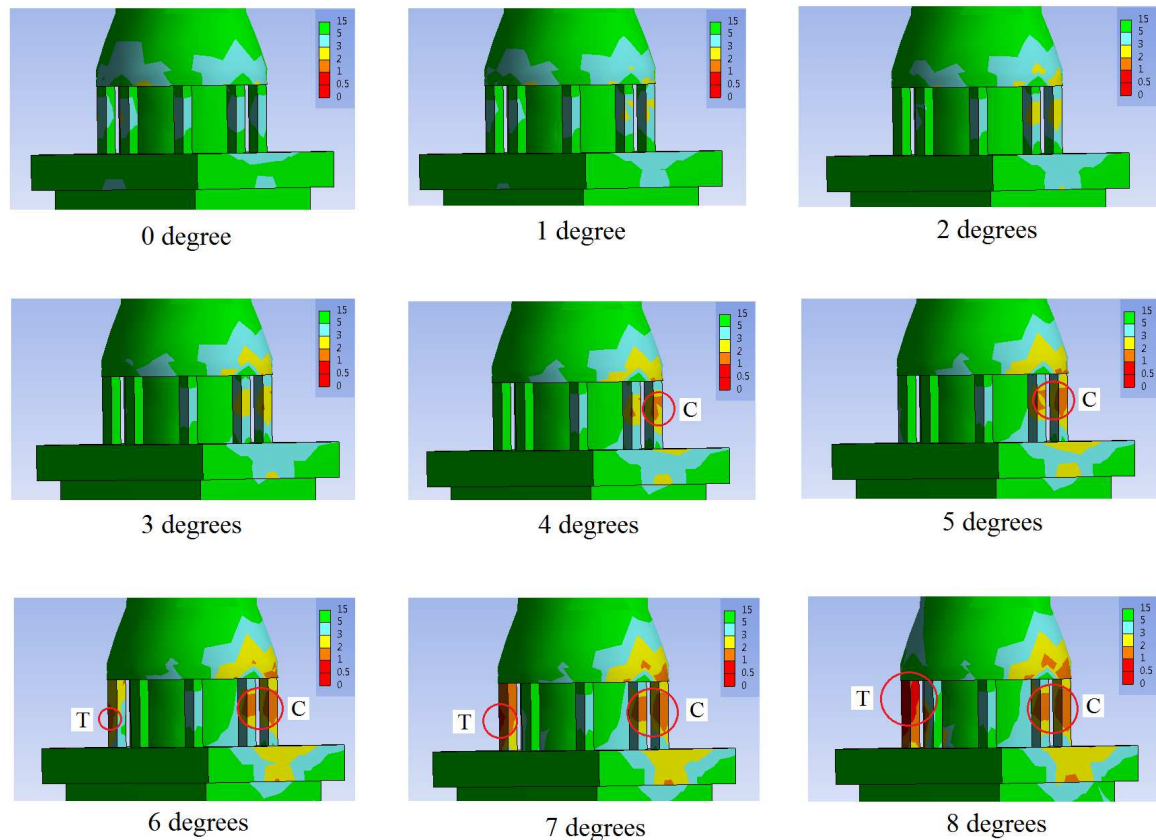


Figure 5.13: Effect of model's tilting on safety factor

5.6 Summary

This section presents the finite element modeling of Chedi Phra Mahathat. The information from the Structure from Motion technique and 3-D laser scan were used to define the geometry of the model. The model was discretized into 4,157 quadratic elements. The elastic modulus was estimated to be 2,200 MPa such that the natural frequency of the model is equal to the measured value at 1.90 Hz. Stress analysis due to self-weight was carried out. The structure is in a safe state with a safety factor greater than 2.0, and bearing stress at base is well below the soil bearing capacity. Modal dynamic properties of the model for the first 30 modes were reported. Effect of assumed tilting angle was also investigated.

References

- [1] ANSYS, Inc., ANSYS workbench 19.2
- [2] Investigation on strength and stability of Chedi Phra Mahathat's foundation. Asian Institute of Technology, Thailand, 1979 (in Thai)
- [3] Archaeological excavation at Wat Phra Mahathat Woramahawihan, Nakhon Si Thammarat, The Fine Arts Department and Nakhon Si Thammarat Province, Thailand, 2018 (in Thai)

- [4] S. Phansuwan, Engineering Analysis of Masonry Historic Monument, Master Thesis, Department of Civil Engineering, Kasetsart University, Thailand, 2000. (in Thai)
- [5] Development of Engineering Database for Assessment and Structural Health Monitoring of Thailand Historic Structures, Department of Civil Engineering, King Mongkut's University of Technology Thonburi, 2018. (in Thai)

Chapter 6 ASSESSMENT OF TRAFFIC INDUCED VIBRATION

6.1 Introduction

Vibration induced by traffic can be harmful for structures if its intensity exceeds certain level. This problem is more pronounced for the case of ancient or historic structures. The characteristics of vibration problem depend on several factors such as properties of soil media in which vibration waves propagate, vehicle types and speed, distance from source. In order to properly investigate the problem, field measurement at the site is necessary. In this section, the results of an assessment of vibration induced by traffic at Wat Phra Mahathat Woramahawihan, Nakhon Si Thammarat are presented. The main objective is to assess the vibration effect on the historical structures in the temple by direct measurement of the vibration intensity. The obtained results were compare with vibration criteria for historic structure. Moreover, an attenuation characteristics of the vibration wave are investigated in this site.

6.2 Method of study

The basis of the study is to directly measure surface wave on ground caused by traffic outside the temple. The main road is Rajdamnern road which run along the east side of the temple. Figure 6.1 shows the location of the vibration sensors and Rajdamnern road. The measuring equipment is the same set which was used in the ambient vibration tests. The sensor arrangement is shown in Figure 6.2 and 6.3, where each sensor has the same spacing. The sensor has 3 components for measurement, however the records clearly show that the vertical component has much higher amplitude than the horizontal components. Therefore, only vertical component is considered in the following discussions. The measurements were conducted during 9 to 12 May 2019.



Figure 6.1 Location of the vibration sensors and Rajdamnern road.

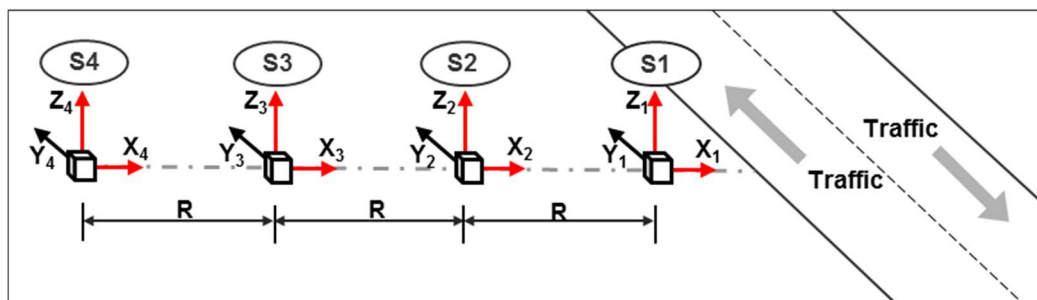


Figure 6.2: Arrangement of vibration sensor (1)



Figure 6.3: Arrangement of vibration sensor (2)

6.3 Site condition

Characteristic of subsoils such as shear wave velocity profile along depth from surface at the site was explored by applying array microtremor measurement. The fundamental concept of array microtremor technique is to simultaneously record the vertical component of microtremors for several positions in the survey area to obtain Rayleigh wave samples propagating from a wide range of azimuthal angles. The coherency spectrum can then be computed for any pair of sensor in the array to evaluate the correlation among them and to determine phase velocity characteristic which is dispersive. From the dispersion relation of phase velocity and frequency, the results from field observations are then compared with those derived theoretically from a horizontally layered earth model by iteration procedure. The results of best-fit shear wave velocity–depth profile can be determined from the inversion analysis. Figure 6.4 show the arrangement of sensor for microtremor array measurement. Figure 6.5 presents the results of this survey as the dispersion curve and the shear wave velocity profile. As shown in Figure 6.5 (b), the shear wave velocity increases along the depth. The average value for the top 30 m (V_{s30}) can be used to classify the dynamic stiffness of subsoils. V_{s30} of this site is 272 m/s which can be considered as stiff soil. For reference, V_{s30} in Amphoe Mueang-Ayutthaya is about 220 m/s, V_{s30} in Amphoe Klong Luang-Pathumthani is about 175 m/s, V_{s30} in Taling Chan district-Bangkok is about 140 m/s and V_{s30} in Amphoe Bangna-Samutprakarn is 90 m/s.



Figure 6.4 Arrangement of sensor for microtremor array measurement

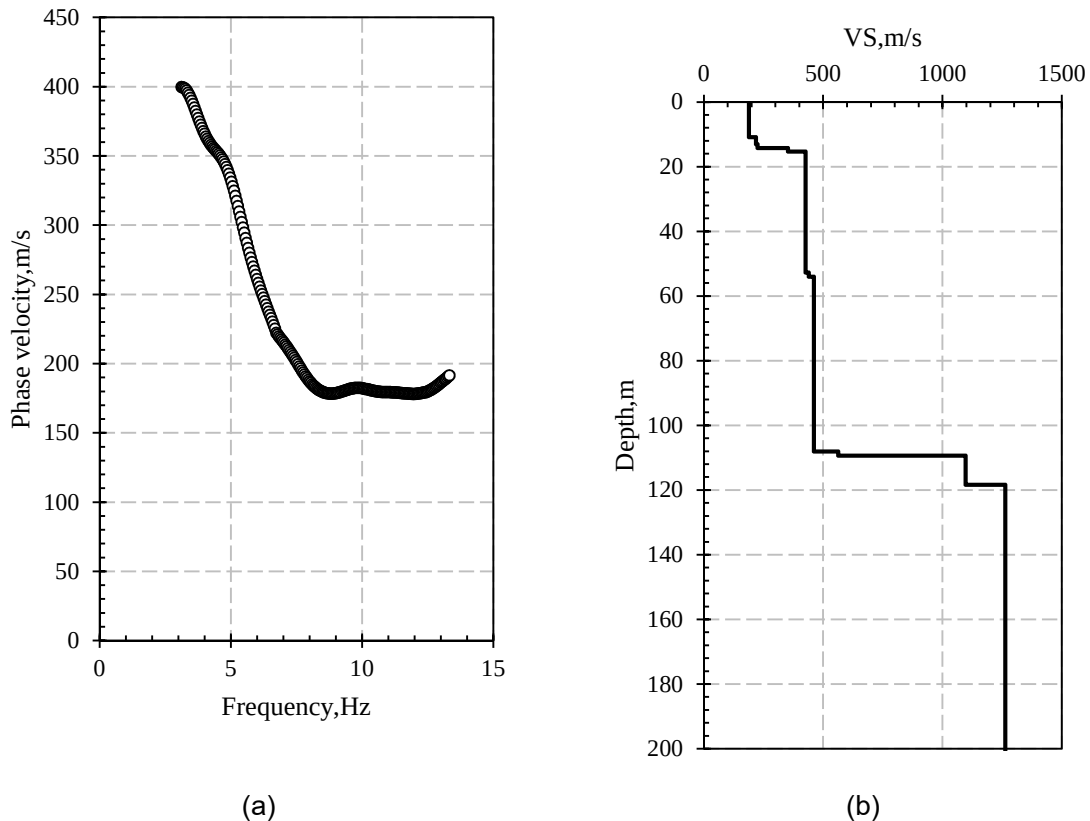


Figure 6.5: (a) Dispersion curve and (b) Shear wave velocity profile at the site

6.4 Attenuation of Vibration Wave

Traffic induces earthborne vibrations which cause stress waves traveling outward from the source in all directions. The propagation characteristics of the wave depends mainly on characteristics of vibration at the source, distance from the source, and soil characteristics. In general, the intensity of the wave decrease with the distance from the source, which is called attenuation characteristic. In this study, attenuation rate of traveling wave from traffic source is investigated by field observation. The attenuation rate was considered as a ratio of the amplitudes between two sensors and the distance between them. In Figure 6.6, for example in the top left figure, the horizontal axis is the amplitude detected at sensor number 1 and the vertical axis is the amplitude detected at sensor number 2. The attenuation rate from these sensors is obtained from the slope of the data points which is simplified as a straight line. This example indicates that the amplitude at the second sensor reduces to be about 78% of the value at the first sensor. The results from others pairs are presented in the same form in this figure. Finally, the attenuation characteristic can be described in Figure 6.7 and 6.8 where the results from this study is compared with the results from other areas having different V_{s30} [1]. The attenuation rate of this area (NST) is similar to Ayutthaya (AYA) as their V_{s30} values are similar. For site with lower V_{s30} , such as Pathumthani (PTE), Bangkok (BKK) and Samutprakarn (SPK), the attenuation rates are slower, or wave can propagate to longer distance than sites in stiffer soil.

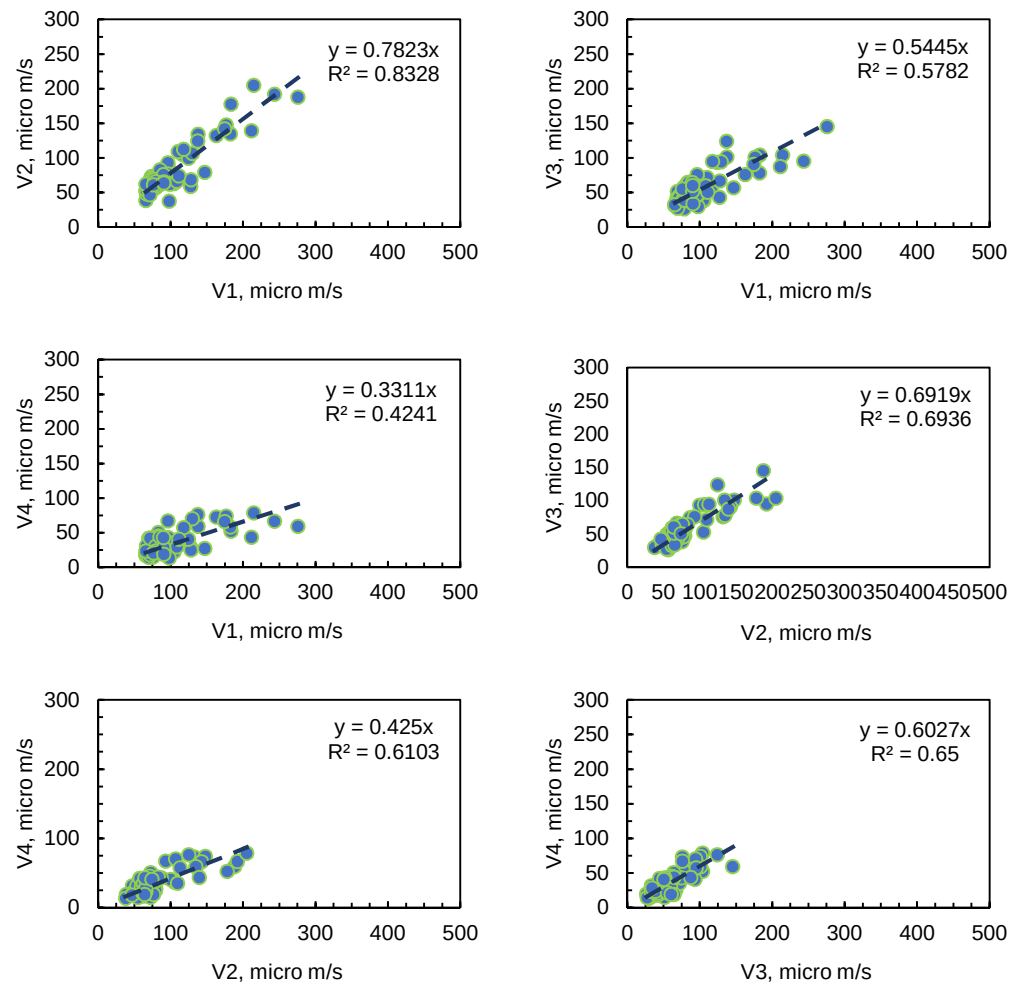


Figure 6.6: Attenuation of vibration amplitude between two sensors.

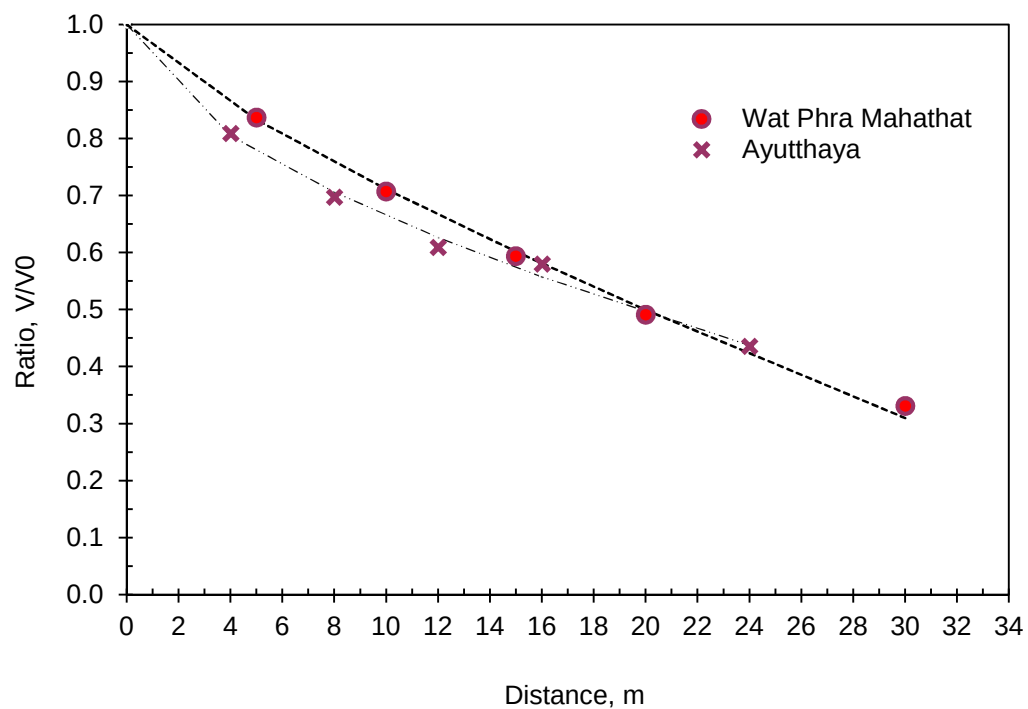


Figure 6.7: Attenuation of vibration and distance from source (1)

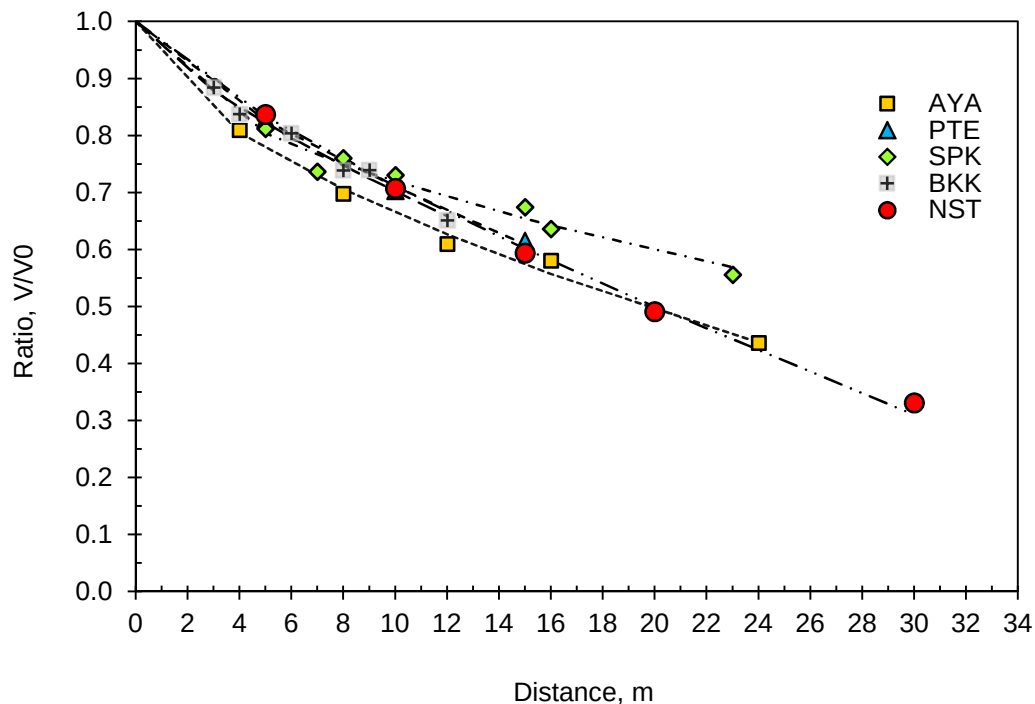


Figure 6.8: Attenuation of vibration and distance from source (2)

6.5 Vibration Intensity and Acceptance Criteria for Historic Buildings

Generally, the recommended values of vibration acceptance criteria for historic buildings are defined in serviceability limit states, which are the limit of hairline or minor cracking, spalling of paint or plaster, excessive deflection or accelerated aging [2]. Most vibration criteria have been developed from experience and are therefore of an empirical in nature. The recommended values depend on type of excitation, frequency content, duration as well as the position where the vibration is considered. The limits for historic buildings are always much lower than those for other types of structures. The tolerable values vary from country to country and no single set of criteria seems to satisfy all requirements. It should be noted that the recommended criteria do not guarantee absence of damage, but reduce its probability of occurrence to acceptably low levels.

In the following, some examples of vibration criteria for historic buildings are reviewed. Bachmann et. al. [2] conducted an extensive review of vibration criteria. Based on his review, there are several recommended vibration limits for historic buildings:

Recommendation A: The maximum vertical (particle) velocity at the building site induced by piling, sheet piling, vibratory compaction and traffic should not exceed

1.8 to 3 mm/s for frequency range of 10 to 30 Hz

1.8 to 5 mm/s for frequency range of 30 to 60 Hz

Recommendation B: The maximum vertical (particle) velocity at the building of great historical value induced by vibratory compactor should not exceed 2 mm/s.

Recommendation C: The maximum vertical velocity for especially sensitive buildings and buildings of cultural and historical value induced by traffic should not exceed 1 mm/s.

In Thailand, the standard on vibration criteria for buildings has been enforced by the Pollution Control Department of Thailand [3]. The DIN 4150-3 standard [4] was used as a model standard for the draft. According to the standard, the maximum velocity of vibrations at the foundation of historic buildings should not exceed 3 mm/s.

In the technical manual of the Departments of the Army and Air force, USA [5], the vibration criterion for buildings is specified as follows: "In order to ensure that the possibility of surficial damage for very old structures is minimized a vibration criterion of 0.05 in/s (1.27 mm/s) is recommended".

In the technical advisory by California Department of Transportation [6], all damage criteria for building are in terms of ground motion at the building's foundations. No allowance is included for the amplifying effects of structural components. For continuous vibrations to which "ruins and ancient monuments" should be subjected, the recommended upper level of peak particle velocity is 2 mm/s.

Based on the above example vibration criteria, the most conservative criterion of 1.0 mm/s will be used as the allowable vibration intensity in this study:

Figure 6.9 (a) shows the maximum vibration intensity along the distance from Rajdamnern road. The data plots are for several tests at different time. The highest peak velocity is 0.58 mm/s at 10 m from the center of the road. As the wave attenuate, the peak velocity is about 0.02 mm/s at 40 m from source. The distance from source to the stupa is more than 40 m., therefore the vibration from traffic in this study is far below the acceptance criteria for historic buildings. Figure 9 (b) shows comparisons with other areas [1].

In order to examine the distribution of vibration intensity, the amplitudes of velocity at the location of sensor 1, which is nearest to the road, are classified into different intensity ranges and plotted against the average rate of occurrence per hour. The result is shown in Figure 6.10 along with the result observed in Ayutthaya. This figure indicates that there is small occurrence rate of high vibration intensity in this site.

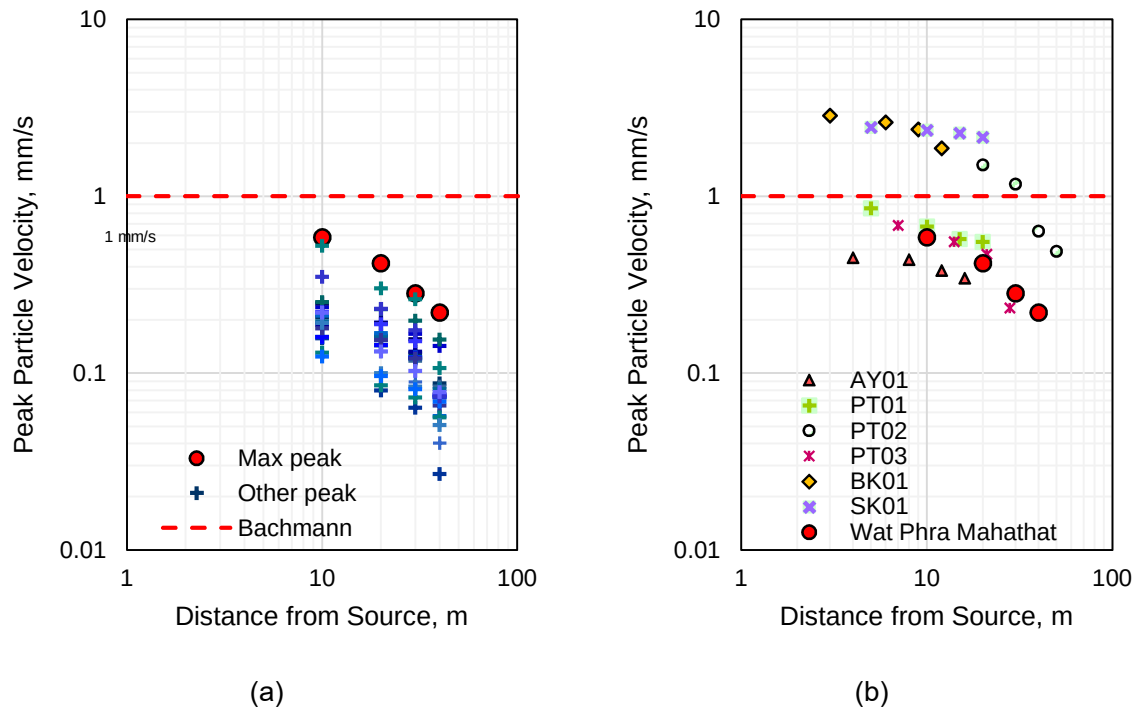


Figure 6.9: Peak particle velocity for all sensors

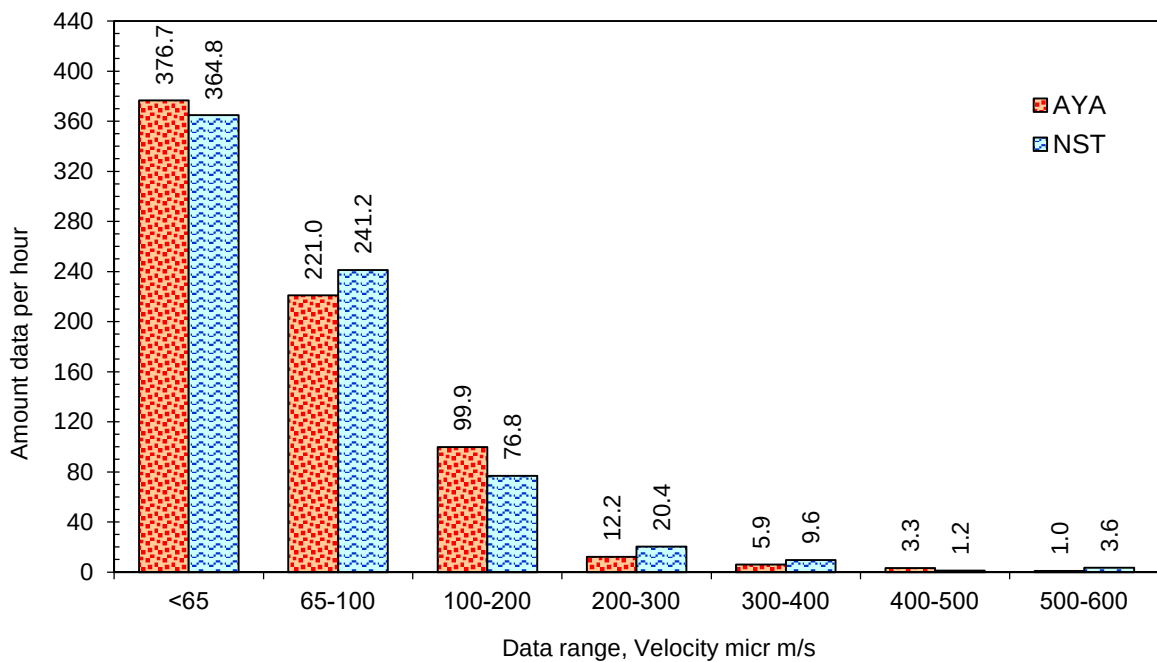


Figure 6.10: Occurrence rate per hour of the peak velocity at sensor number 1

6.6 Reference

- [1] Choopthong W. and Poovarodom N., (2018) "Assessment of Traffic-Induced Vibration Effects on Historical Structures in Ayutthaya" Thai Science and Technology Journal, Vol. 26, No. 6, Pages 1075-1085.
- [2] Bachmann, H. et. al., (1995) Vibration problems in structures; practical guidelines, Birkhauser.

- [3] Pollution Control Department, (2010) Standard for vibration effects on buildings and components, Bangkok.
- [4] Deutsches Institut für Normung, (1999) DIN 4150-3, Structural Vibration Part 3; Effects of Vibration on Structures.
- [5] Departments of the Army and Air force, Noise and vibration control, (1995) Technical report Army TM 5-805-4, Air Force AFJMAN 32-1090.
- [6] California Department of Transportation, (2002) Transportation related earthborne vibrations, Technical advisory, vibration, TAV-02-01-R9601.

Chapter 7 SUMMARY

In this study, engineering measures were applied for documentation and assessment of the existing conditions of Wat Phra Mahathat Woramahawihan, Nakhon Si Thammarat. The major outputs from the study can be summarized as follows.

1. Terrestrial laser scanning technology was used to collect point cloud data for 3D digital documentation. In this study, seventy-four laser scanner stations were used to collect the point cloud data of the area, and approximately one billion 3D point clouds of the Wat Phra Mahathat Woramahawihan were generated. Finally, point cloud data are stored and shared using Webshare application and they are accessible for monitoring in the future conservation plan.
2. 3D digital documentation was also conducted by the image-based photogrammetry technique to estimate the tilt angle and inspect the surface damage. The 3D model was also created for Finite Element modeling to evaluate stability of the structure. By comparing with the laser scanner, the 3D model from the image-based technique provides an identical result. It was also found that the top of the stupa was tilted by about 1.5 degree southward. This technique also provides a practical measure for inspection of surface damage.
3. Ambient vibration tests were conducted to identify the natural frequencies of Chedi Phra Mahathat and 22 subsidiary chedis. The identified natural frequency for the first mode of vibration of Chedi Phra Mahathat is about 1.96 Hz. and 1.84 Hz. In X and Y direction, respectively. The satellite chedi has the natural frequency for the first mode of vibration of about 10.5 Hz. The results of the identified natural periods of the fundamental mode and height of the subsidiary chedis are presented as the natural periods increase with height of the chedis.
4. Finite element modeling of Chedi Phra Mahathat was studied in order to perform structural analysis due to self-weight. The information from the Structure from Motion technique and 3-D laser scan were used to define the geometry of the model. The elastic modulus was estimated to be 2,200 MPa such that the natural frequency of the model is equal to the measured value at 1.90 Hz. The structure is in safe state with safety factor greater than 2.0, and bearing stress at base is well below the soil bearing capacity. Modal dynamic properties of the model for the first 30 modes were reported. Effect of assumed tilting angle was also investigated.

5. Vibration induced from traffic was studied by field measurement in order to assess its effects to the historical site. Site characteristic was firstly determined by microtremor observation, and the obtained average shear wave velocity from surface to 30 m depth was 272 m/s. Attenuation of vibration amplitude was examined and it was found that due to stiff subsoils the attenuation rate is faster than other soft soil areas. The highest vibration velocity found in this study was lower than the recommendation provided in the international standards of vibration effect on historical structure.